

Analysis II

1 Uniform convergence

Let $E \subseteq \mathbb{R}$ and $f_n : E \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be a sequence of functions. What could it mean for the sequence (f_n) to converge? What if E is some arbitrary set?

The most basic notion would be for any $x \in E$, we require the sequence of real numbers $(f_n(x))$ to converge. If this holds, we can define a new function $f : E \rightarrow \mathbb{R}$ by simply setting $f(x) = \lim_{n \rightarrow \infty} f_n(x)$.

Definition 1.1 – Pointwise convergence

Let $f_n : E \rightarrow \mathbb{R}$ be a sequence of functions. (f_n) converges to f pointwise if

$$f(x) = \lim_{n \rightarrow \infty} f_n(x)$$

and we write $f_n \rightarrow f$ pointwise on E .

Are useful properties such as continuity, differentiability or integrability preserved in the limit? For example, if f_n is continuous on E for all n , does it follow that f is continuous on E ?

Example 1.1

No. Let $E = [-1, 1]$ and $f_n(x) = x^{1/(2n+1)}$, then

$$f_n(x) \rightarrow f(x) = \begin{cases} 1 & \text{if } 0 < x \leq 1 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } -1 \leq x < 0 \end{cases}$$

which is not continuous.

Example 1.2

Integrability is not preserved either. Let $E = [0, 1]$ and $q_1, q_2, q_3, \dots$ be an enumeration of the rationals on $[0, 1]$. Define

$$f_n(x) = \begin{cases} 1 & \text{if } x \in \{q_1, \dots, q_n\} \\ 0 & \text{otherwise} \end{cases}$$

which is integrable on $[0, 1]$ for all n since it is continuous everywhere except on the finite set $\{q_1, \dots, q_n\}$. However,

$$f_n(x) \rightarrow f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

which is not integrable.

Example 1.3

If $f_n \rightarrow f$ pointwise and f_n, f are integrable, does it follow that $\int_E f_n \rightarrow \int_E f$? No. Let $E = [0, 2]$ and

$$f_n(x) = \begin{cases} n^2 x & \text{if } 0 \leq x \leq 1/n \\ n^2(2/n - x) & \text{if } 1/n \leq x \leq 2/n \\ 0 & \text{otherwise} \end{cases}$$

then $f_n(x) \rightarrow 0$ for all $x \in E$. f_n and f are continuous and hence integrable, with $\int_E f_n = 1$ but $\int_E f = 0$, so $\int_E f_n \not\rightarrow \int_E f$.

We need a stronger notion of convergence for sequences (f_n) to ensure that properties such as continuity, integrability, etc. are preserved. It can't be too strong however, for example $f_n \rightarrow f$ meaning $f_n = f$ for all sufficiently large n is a useless notion. A good middle ground is uniform convergence.

Definition 1.2 – Uniform convergence

Let $f_n, f : E \rightarrow \mathbb{R}$. We say (f_n) converges to f uniformly on E if the following holds:

$$\forall \varepsilon > 0, \exists N = N(\varepsilon) \text{ such that for every } n \geq N \text{ and for every } x \in E, \text{ we have } |f_n(x) - f(x)| < \varepsilon.$$

This is equivalent to

$$\forall \varepsilon > 0, \exists N = N(\varepsilon) \text{ such that for all } n \geq N, \text{ we have } \sup_{x \in E} |f_n(x) - f(x)| < \varepsilon.$$

Or, $\sup_E |f_n - f| \rightarrow 0$ as $n \rightarrow \infty$.

As a comparison, pointwise convergence means $\forall x \in E, \forall \varepsilon > 0, \exists N = N(\varepsilon, x)$ such that for all $n \geq N$, we have $|f_n(x) - f(x)| < \varepsilon$. So for uniform convergence, the same N works for all x (given an ε). For pointwise convergence, N is allowed to depend on x .

Notice that if $f_n \rightarrow f$ uniformly on E , then $f_n \rightarrow f$ pointwise on E . This gives a useful way to compute uniform limits of sequences of functions. First, check if the pointwise limit exists. If not, then there's no uniform limit. If it exists, then we can check whether $\sup_E |f_n - f| \rightarrow 0$.

Definition 1.3 – Uniformly Cauchy

Let $f_n : E \rightarrow \mathbb{R}$ be a sequence of functions. We say (f_n) is uniformly Cauchy if

$$\forall \varepsilon > 0, \exists N = N(\varepsilon) \text{ such that for all } n, m \geq N, \text{ we have } \sup_{x \in E} |f_n(x) - f_m(x)| < \varepsilon.$$

Theorem 1.1 – Cauchy's criterion for uniform convergence

Let $f_n : E \rightarrow \mathbb{R}$ be a sequence of functions, then the following equivalence holds:

$$(f_n) \text{ converges uniformly on } E \text{ if and only if } (f_n) \text{ is uniformly Cauchy on } E.$$

Proof. If (f_n) converges uniformly to $f : E \rightarrow \mathbb{R}$, then $\forall \varepsilon > 0, \exists N = N(\varepsilon)$ such that $\sup_E |f_n - f| < \varepsilon$ for all $n \geq N$. By the triangle inequality, $\forall x \in E$,

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq |f_n(x) - f(x)| + |f_m(x) - f(x)| \\ &\leq \sup_E |f_n(x) - f(x)| + \sup_E |f_m(x) - f(x)| \end{aligned}$$

So, by taking supremum, for $n, m \geq N$, $\sup_E |f_n - f_m| < 2\varepsilon$. Since ε is arbitrary, this shows that (f_n) is

uniformly Cauchy.

Conversely, suppose (f_n) is uniformly Cauchy, then the sequence of real numbers $(f_n(x))$ is Cauchy for all $x \in E$, and so it has a limit. Calling this limit $f(x)$, we have $f_n \rightarrow f$ pointwise on E . To check that $f_n \rightarrow f$ uniformly on E , pick any $\varepsilon > 0$, and note that by the hypothesis that (f_n) is uniformly Cauchy, $\exists N = N(\varepsilon)$ such that $n, m \geq N$ implies $|f_n(x) - f_m(x)| < \varepsilon$ for all $x \in E$. Fixing $n \geq N$ and letting $m \rightarrow \infty$, we get $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$. So, $\sup_{x \in E} |f_n(x) - f(x)| \leq \varepsilon$, that is $f_n \rightarrow f$ uniformly on E . $\square$

Revisiting Example 1.3, $f_n \rightarrow 0$ pointwise on $[0, 2]$ but $\sup_{[0,2]} |f_n - 0| = n$, so $f_n \not\rightarrow f$ uniformly on $[0, 2]$.

Example 1.4

A function's domain could change whether it converges uniformly. Take $f_n : \mathbb{R} \rightarrow \mathbb{R}$ with $f_n(x) = x/n$. $f_n \rightarrow 0$ pointwise on $\mathbb{R}$, but $\sup_{\mathbb{R}} |f_n - 0| = \infty$, so $f_n \not\rightarrow 0$ uniformly on $\mathbb{R}$. However, if we fix $a \in (0, \infty)$ and let $f_n : [-a, a] \rightarrow \mathbb{R}$ with $f_n(x) = x/n$ as before, $\sup_{[-a,a]} |f_n - 0| = a/n \rightarrow 0$, so $f_n \rightarrow 0$ uniformly on $[-a, a]$.

Theorem 1.2 – Continuity is preserved under uniform convergence

Let $f_n, f : [a, b] \rightarrow \mathbb{R}$ and suppose that $f_n \rightarrow f$ uniformly on $[a, b]$. If $x \in [a, b]$ is such that f_n is continuous at x for all n , then f is continuous at x .

Proof. Let $\varepsilon > 0$. By the uniform convergence of $f_n \rightarrow f$, there exists $N = N(\varepsilon)$ such that $\sup_{y \in [a,b]} |f_n(y) - f(y)| < \varepsilon$ for all $n \geq N$. By continuity of f_N at x , we have $\delta = \delta(N, x, \varepsilon) > 0$ such that if $y \in [a, b]$ and $|x - y| < \delta$, we have $|f_N(y) - f_N(x)| < \varepsilon$. Then, if $y \in [a, b]$ and $|x - y| < \delta$, we have

$$\begin{aligned} |f(y) - f(x)| &\leq |f(y) - f_N(y)| + |f_N(y) - f_N(x)| + |f_N(x) - f(x)| \\ &< 3\varepsilon \end{aligned}$$

Since ε is arbitrary, f is continuous at x . $\square$

Applying this theorem to every point in an interval, we see that

Corollary 1.1 – Uniform limits of continuous functions are continuous

If $f_n, f : [a, b] \rightarrow \mathbb{R}$ and $f_n \rightarrow f$ uniformly on $[a, b]$ and f_n is continuous on $[a, b]$ for each n , then f is continuous on $[a, b]$.

Let $C([a, b]) = \{f : [a, b] \rightarrow \mathbb{R} \mid f \text{ is continuous on } [a, b]\}$ be the space of continuous functions on $[a, b]$. Another way to state Corollary 1.1 would be

Theorem 1.3

Let (f_n) be a uniformly Cauchy sequence of functions in $C([a, b])$, then (f_n) to a function $f \in C([a, b])$.

Theorem 1.4 – Uniform convergence implies convergence of integrals

Let $f_n, f : [a, b] \rightarrow \mathbb{R}$ be functions such that f_n, f are bounded and integrable on $[a, b]$. If $f_n \rightarrow f$ uniformly on $[a, b]$, then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$$

The assumption that f is integrable is redundant. We will see later that the integrability of f_n and the uniform convergence of $f_n \rightarrow f$ implies that f is integrable.

Proof.

$$\begin{aligned} \left| \int_a^b f_n(x) dx - \int_a^b f(x) dx \right| &= \left| \int_a^b (f_n(x) - f(x)) dx \right| \\ &\leq \int_a^b |f_n(x) - f(x)| dx \\ &\leq \sup_{x \in [a,b]} |f_n(x) - f(x)| (b - a) \rightarrow 0 \end{aligned}$$

□

The connection between differentiation and uniform convergence is more subtle. If $f_n \rightarrow f$ uniformly on some interval, and if f_n are differentiable on that interval for all n , it does not follow that f is differentiable. Even if f is differentiable, $f'_n(x) \rightarrow f'(x)$ might be false.

Example 1.5

To see this, let $f_n : [-1, 1] \rightarrow \mathbb{R}$ with $f_n(x) = |x|^{1+1/n}$.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f_n(x) - f_n(0)}{x} &= \lim_{x \rightarrow 0} \frac{f_n(x)}{x} \\ &= \lim_{x \rightarrow 0} \operatorname{sgn}(x) x^{1/n} = 0 \end{aligned}$$

so f_n is differentiable at 0 with $f'_n(0) = 0$. It is clear that f_n is differentiable if $x \neq 0$. If $f(x) = |x|$, then $f_n \rightarrow f$ uniformly on $[-1, 1]$, but f is not differentiable at $x = 0$.

Example 1.6

Consider $f_n : \mathbb{R} \rightarrow \mathbb{R}$ with $f_n(x) = \sin(nx) / \sqrt{n}$. $f_n \rightarrow 0$ uniformly on $\mathbb{R}$ since $\sup_{\mathbb{R}} |f_n(x)| \leq 1/\sqrt{n} \rightarrow 0$. So, we have a differentiable limit. However, $f'_n(x) = \sqrt{n} \cos(nx)$, so $f'_n(0) \rightarrow \infty$ and $f'_n(x) \not\rightarrow f'(x)$ pointwise on $\mathbb{R}$.

Theorem 1.5

Let $f_n : [a, b] \rightarrow \mathbb{R}$ be a sequence of differentiable functions (at the end points, the one-sided derivative exists). Suppose $f'_n \rightarrow g$ uniformly on $[a, b]$ for some function $g : [a, b] \rightarrow \mathbb{R}$ and for some $c \in [a, b]$, the sequence $(f_n(c))$ converges. Then,

- (i) (f_n) converges uniformly to some function $f : [a, b] \rightarrow \mathbb{R}$;
- (ii) f is differentiable on $[a, b]$ and $f'(x) = g(x)$ for all $x \in [a, b]$, that is

$$\left[\lim_{n \rightarrow \infty} f_n \right]' = \lim_{n \rightarrow \infty} f'_n.$$

So, given the assumptions, we can interchange differentiation with taking limits. If we assume that f'_n are continuous, then the proof is more straightforward and can be based on the fundamental theorem of calculus.

Proof.

- (i) By the mean value theorem applied to $f_n - f_m$, we have that for any $x \in [a, b]$,

$$f_n(x) - f_m(x) = f_n(c) - f_m(c) + (x - c)(f'_n - f'_m)(x_{n,m})$$

for some $x_{n,m}$ between x and c so

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq |f_n(c) - f_m(c)| + (b-a)|f'_n(x_{n,m}) - f'_m(x_{n,m})| \\ &\leq |f_n(c) - f_m(c)| + (b-a)\sup(f'_n - f'_m) \rightarrow 0 \end{aligned}$$

as $n, m \rightarrow \infty$ by our assumptions, so (f_n) is uniformly Cauchy, and hence there is a function $f : [a, b] \rightarrow \mathbb{R}$ such that $f_n \rightarrow f$ uniformly.

(ii) Fix some arbitrary $y \in [a, b]$. Define

$$h(x) = \begin{cases} \frac{f(x) - f(y)}{x - y} & \text{if } x \neq y \\ g(y) & \text{if } x = y. \end{cases}$$

So the differentiability of f at y with $f'(y) = g(y)$ is equivalent to the continuity of h at y . Let

$$h_n(x) = \begin{cases} \frac{f_n(x) - f_n(y)}{x - y} & \text{if } x \neq y \\ f'_n(y) & \text{if } x = y. \end{cases}$$

Then, since f_n is differentiable at y , we see that h_n is continuous on $[a, b]$. The pointwise limit of (h_n) is h since $f'_n \rightarrow g$ at $x = y$ by assumption. If we can show that (h_n) converges uniformly on $[a, b]$, then the limit must be h since it is the pointwise limit, and the limit h must be continuous by Corollary 1.1. To check that (h_n) converges uniformly, we check that (h_n) is uniformly Cauchy,

$$h_n(x) - h_m(x) = \begin{cases} \frac{(f_n - f_m)(x) - (f_n - f_m)(y)}{x - y} & \text{if } x \neq y \\ (f'_n - f'_m)(y) & \text{if } x = y. \end{cases}$$

By the mean value theorem,

$$h_n(x) - h_m(x) = \begin{cases} (f_n - f_m)'(x_{n,m}) & \text{if } x \neq y \\ (f_n - f_m)'(y) & \text{if } x = y \end{cases}$$

for some $x_{n,m}$ between x and y , so

$$\sup_{[a,b]} |h_n - h_m| \leq \sup_{[a,b]} |f'_n - f'_m| \rightarrow 0$$

as $n, m \rightarrow \infty$, that is (h_n) is uniformly Cauchy and we are done. □

Example 1.7

Note that (f'_n) need not be continuous. Consider $f : [-1, 1] \rightarrow \mathbb{R}$ with

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

f is differentiable on $[-1, 1]$ with

$$f'(x) = \begin{cases} 2x \sin(1/x) - \cos(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

which is not continuous at $x = 0$. We can then take $f_n(x) = f(x)$ for all n .

If we assume that (f'_n) are continuous in addition to all the other hypothesis, then we have a shorter proof of Theorem 1.5.

Proof. For any $x \in [a, b]$, then

$$f_n(x) = f_n(c) + \int_c^x f'_n(t) dt$$

by the fundamental theorem of calculus. So,

$$\begin{aligned} |f_n(x) - f_m(x)| &= \left| (f_n(c) - f_m(c)) + \int_c^x (f'_n(t) - f'_m(t)) dt \right| \\ &\leq |f_n(c) - f_m(c)| + \sup_{t \in [a, b]} |f'_n(t) - f'_m(t)| (b - a) \rightarrow 0 \end{aligned}$$

as $n, m \rightarrow \infty$. Hence, (f_n) is uniformly Cauchy and so converges uniformly to some $f : [a, b] \rightarrow \mathbb{R}$.

Note that

$$\int_c^x f'_n(t) dt \rightarrow \int_c^x g(t) dt.$$

This is by the uniform convergence $f'_n \rightarrow g$ which also implies g is continuous hence integrable. So,

$$f_n(x) = f_n(c) + \int_c^x f'_n(t) dt \xrightarrow{n \rightarrow \infty} f(x) = f(c) + \int_c^x g(t) dt.$$

By the fundamental theorem of calculus again, f is differentiable with

$$f'(x) = g(x) = \lim_{n \rightarrow \infty} f'_n(x).$$

□

Proposition 1.1

Let $f_n, g_n : E \rightarrow \mathbb{R}$. If $f_n \rightarrow f$ uniformly on E and $g_n \rightarrow g$ uniformly on E , then

- (i) $f_n + g_n \rightarrow f + g$ uniformly on E ;
- (ii) if $h : E \rightarrow \mathbb{R}$ is a bounded function, then $hf_n \rightarrow hf$ uniformly on E .

Example 1.8

Note that (ii) is false without the boundedness assumption on h . Let $f_n, h : \mathbb{R} \rightarrow \mathbb{R}$ with $f_n(x) = 1/n$ and $h(x) = x$. $f_n \rightarrow 0$ uniformly on $\mathbb{R}$ but $(hf_n)(x) = x/n \not\rightarrow 0$ uniformly on $\mathbb{R}$. Or, on an interval, let $f_n, h : (0, 1) \rightarrow \mathbb{R}$ with $f_n(x) = 1/n$ and $h(x) = 1/x$.

Definition 1.4 – Convergence of series of functions

Let $g_n : E \rightarrow \mathbb{R}$. Write

$$f_n = \sum_{j=1}^n g_j \text{ defined by } f_n(x) = \sum_{j=1}^n g_j(x),$$

so the partial sums $f_n : E \rightarrow \mathbb{R}$. Then, we say the series of functions $\sum_{n=1}^{\infty} g_n$

- (i) converges at a point $x \in E$ if the sequence $(f_n(x))$ converges;
- (ii) converges uniformly on E if the sequence (f_n) converges uniformly on E ;
- (iii) converges absolutely at $x \in E$ if the series $\sum_{n=1}^{\infty} |g_n(x)|$ converges;
- (iv) converges absolutely uniformly on E if the series $\sum_{n=1}^{\infty} |g_n|$ converges uniformly on E .

We know from Analysis I that absolute convergence implies convergence for series of numbers. From this, we know that if $\sum_{n=1}^{\infty} g_n$ converges absolutely at a point $x \in E$, then $\sum_{n=1}^{\infty} g_n$ converges at x . Similarly, we have

Proposition 1.2 – Absolute uniform convergence implies uniform convergence

If $g_n : E \rightarrow \mathbb{R}$, and if $\sum_{n=1}^{\infty} g_n$ converges absolutely uniformly on E , then $\sum_{n=1}^{\infty} g_n$ converges uniformly on E .

Proof. Let $f_n = \sum_{j=1}^{\infty} g_j$. Assuming $n > m$,

$$\begin{aligned} |f_n(x) - f_m(x)| &= \left| \sum_{j=m+1}^n g_j(x) \right| \\ &\leq \sum_{j=m+1}^n |g_j(x)| \\ &= h_n(x) - h_m(x) \text{ where } h_n(x) = \sum_{j=1}^n |g_j(x)|. \end{aligned}$$

Taking the supremum,

$$\sup_{x \in E} |f_n(x) - f_m(x)| \leq \sup_{x \in E} |h_n(x) - h_m(x)| \rightarrow 0$$

as $n, m \rightarrow \infty$ by the hypothesis, so (f_n) converges uniformly on E . $\square$

Note that uniform convergence with absolute convergence at every point does not imply absolute uniform convergence.

Example 1.9

Let $g_n : [0, 1) \rightarrow \mathbb{R}$ with $g_n(x) = (-1)^n x^n / n$. This converges absolutely for all $x \in [0, 1)$. Also, the series converges uniformly on $[0, 1)$. But, this does not converge absolutely uniformly.

$$\sum_{j=1}^n |g_j(x)| - \sum_{j=1}^m |g_j(x)| = \sum_{j=m+1}^n \frac{x^j}{j} \geq \left(\frac{1}{m+1} + \cdots + \frac{1}{n} \right) x^n.$$

Since

$$\sup_{x \in [0, 1)} \left(\sum_{j=m+1}^n \frac{1}{j} \right) x^n = \sum_{j=m+1}^n \frac{1}{j} \rightarrow \infty$$

as $n \rightarrow \infty$ for all fixed m , $\sum_{j=1}^n |g_j(x)|$ does not converge uniformly.

Theorem 1.6 – Weierstrass M-test

Let $g_n : E \rightarrow \mathbb{R}$. Suppose that there exists M_n such that

$$\sup_{x \in E} |g_n(x)| \leq M_n \text{ and } \sum_{n=1}^{\infty} M_n \text{ converges.}$$

Then, $\sum_{n=1}^{\infty} g_n$ converges absolutely uniformly on E .

Proof. Let $h_n(x) = \sum_{j=1}^n |g_j(x)|$. For $n > m$,

$$\begin{aligned} h_n(x) - h_m(x) &= \sum_{j=m+1}^n |g_j(x)| \\ &\leq \sum_{j=m+1}^n M_j = \sum_{j=1}^n M_j - \sum_{j=1}^m M_j. \end{aligned}$$

So,

$$\sup_{x \in E} |h_n(x) - h_m(x)| \leq \left| \sum_{j=1}^n M_j - \sum_{j=1}^m M_j \right| \rightarrow 0$$

as $n, m \rightarrow \infty$. Hence, (h_n) is uniformly Cauchy and so converges uniformly. $\square$

We will now specialise the preceding discussion to the case $g_n(x) = c_n(x - a)^n$ for fixed $c_n, a \in \mathbb{R}$. We get the real power series

$$\sum_{n=0}^{\infty} c_n(x - a)^n.$$

Theorem 1.7 – Radius of convergence

Let $\sum_{n=0}^{\infty} c_n(x - a)^n$ be a real power series. Then, there exists $R \in [0, \infty]$, called the radius of convergence of the power series, such that

- (i) if $|x - a| < R$, then the power series converges absolutely at x ;
- (ii) if $|x - a| > R$, then the power series diverges;
- (iii) the radius of convergence is given by

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |c_n|^{1/n},$$

where we set $1/0 = \infty$ and $1/\infty = 0$;

- (iv) for any $r \in (0, R)$, we have that the power series converges absolutely uniformly on $[a - r, a + r]$. In particular, the function $f : (a - R, a + R) \rightarrow \mathbb{R}$ with $f(x) = \sum_{n=0}^{\infty} c_n(x - a)^n$ is continuous on $(a - R, a + R)$.

Before proving this, an aside on the limit superior/inferior. If (a_n) be a sequence of real numbers, let $p_n = \sup \{a_m : m \geq n\}$ and $q_n = \inf \{a_m : m \geq n\}$. We define

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} p_n \text{ and } \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} q_n.$$

These exist in $\mathbb{R} \cup \{\pm\infty\}$ since (p_n) and (q_n) are monotone. Also, we have that

- (i) $\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$
- (ii) $\liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = l \iff \lim_{n \rightarrow \infty} a_n = l.$

Proof. Parts (i) to (iii) were done in IA Analysis I.

For (iv), note first that the power series converges absolutely at $x = a + r$, that is $\sum_{n=0}^{\infty} |c_n| r^n$ is convergent. Since $|c_n(x - a)^n| \leq |c_n| r^n$ for any $x \in [a - r, a + r]$, we can apply the Weierstrass M-test with $M_n = |c_n| r^n$ to conclude that $\sum_{n=0}^{\infty} c_n(x - a)^n$ converges absolutely uniformly on $[a - r, a + r]$. Since the uniform limit of continuous functions is continuous, it follows that $f(x) = \sum_{n=0}^{\infty} c_n(x - a)^n$ is continuous on $[a - r, a + r]$. As $0 < r < R$ is arbitrary, f is continuous at any $x \in (a - R, a + R)$. $\square$

If the power series $\sum_{n=0}^{\infty} c_n(x - a)^n$ has radius of convergence $R > 0$, then the power series is said to converge locally uniformly on the interval of convergence $(a - R, a + R)$, that is it converges uniformly on $[a - r, a + r]$ for any $r \in (0, R)$.

Let $f(x) = \sum_{n=0}^{\infty} c_n(x - a)^n$ be a power series having radius of convergence R with $0 < R < \infty$. If the power series converges at one of the boundary points of the interval of convergence, say at $x = a + R$, that is $\sum_{n=0}^{\infty} c_n R^n$ is convergent, then $\lim_{x \rightarrow (a+R)^-} f(x) = \sum_{n=0}^{\infty} c_n R^n$, so f extends to $(a - R, a + R]$ as a continuous

function with $f(a+R) = \sum_{n=0}^{\infty} c_n R^n$. Moreover, under the same condition that $\sum_{n=0}^{\infty} c_n R^n$ converges, we have the series converges uniformly on $[a-r, a+R]$ for any $r \in (0, R)$. The same discussion applies at the end point $a-R$.

Theorem 1.8 – Differentiation of power series

Let $\sum_{n=0}^{\infty} c_n(x-a)^n$ be a power series with radius of convergence $R > 0$. Let $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$, so $f : (a-R, a+R) \rightarrow \mathbb{R}$. We have the following.

- (i) The derived series $\sum_{n=1}^{\infty} n c_n(x-a)^{n-1}$ has the same radius of convergence R .
- (ii) f is differentiable on $(a-R, a+R)$ with $f'(x) = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1}$ for all $x \in (a-R, a+R)$.

Proof.

- (i) $\limsup_{n \rightarrow \infty} (n|c_n|)^{1/n} = \limsup_{n \rightarrow \infty} |c_n|^{1/n}$ since $\lim_{n \rightarrow \infty} n^{1/n} = 1$.
- (ii) Let $f_n(x) = \sum_{j=0}^n c_j(x-a)^j$. These are differentiable on $\mathbb{R}$, with $f'_n(x) = \sum_{j=1}^n j c_j(x-a)^{j-1}$. By (i), we know that (f'_n) converges uniformly on $[a-r, a+r]$ for any $r \in (0, R)$, and $f_n(a) = c_0$ for n so $(f_n(a))$ is convergent. Hence, f is differentiable on $[a-r, a+r]$ with $f'(x) = \lim_{n \rightarrow \infty} f'_n(x) = \sum_{j=1}^{\infty} j c_j(x-a)^{j-1}$. Again, since r is arbitrary, we obtain the result. □

By repeatedly applying this theorem, we get that if $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$, then $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ with $x \in (a-R, a+R)$ is differentiable to any order k on the same interval $(a-R, a+R)$. The k -th derivative is

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\dots(n-k+1)c_n(x-a)^{n-k}$$

with $x \in (a-R, a+R)$. Taking $x = a$, we get $c_k = f^{(k)}(a)/k!$. This says that f is uniquely determined by its values in an arbitrarily small interval around the point $x = a$, since that is all we used to compute the derivative $f^{(k)}(a)$.

2 Uniform continuity and Riemann integrability

Definition 2.1 – Uniform continuity

Let $E \subseteq \mathbb{R}$ and $f : E \rightarrow \mathbb{R}$. We say f is uniformly continuous on E if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } x, y \in E \text{ and } |x - y| < \delta \text{ implies } |f(x) - f(y)| < \varepsilon.$$

Contrast this with the definition of continuity at a point $x \in E$. For continuity at $x \in E$, we require that for all $\varepsilon > 0$, there exists $\delta > 0$ (possibly depending on both ε and x) such that for any $y \in E$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$. For uniform continuity, we require a δ that does not depend on where x is in E .

Example 2.1

Clearly, uniform continuity implies continuity at each $x \in E$. The converse is false. Take $f(x) = 1/x$ with $x \in (0, 1)$. This is continuous at each $x \in (0, 1)$. For any $\delta > 0$, if $x \in (0, \delta)$ and with $y = x/2$, $|x - y| = x/2 < \delta$, but $|f(x) - f(y)| = |1/x - 2/x| = 1/x > 1$. So no choice of $\delta > 0$ for $\varepsilon = 1$ satisfies the definition.

Example 2.2

Even if a continuous function f is bounded on its domain, it need not be uniformly continuous. Take $f(x) = \sin(1/x)$ with $x \in (0, 1)$. Take $x_n = 1/(2n\pi)$ and $y_n = 1/((2n + 1/2)\pi)$. Then, $|x_n - y_n| \rightarrow 0$ as $n \rightarrow \infty$ but $|f(x_n) - f(y_n)| = 1$ for all n .

Theorem 2.1

Let $[a, b]$ be a closed and bounded interval, and $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then, f is uniformly continuous.

Proof. We argue by contradiction. If f is not uniformly continuous on $[a, b]$, then there exists $\varepsilon > 0$ for which no δ works. That is, we can take $\delta = 1/n$ and get $x_n, y_n \in [a, b]$ with $|x_n - y_n| < 1/n$ and $|f(x_n) - f(y_n)| \geq \varepsilon$. By the Bolzano-Weierstrass theorem, the sequence (x_n) has a subsequence $(x_{n_k})_k$ that converges to a point $x \in [a, b]$.

$$\begin{aligned} |x - y_{n_k}| &\leq |x - x_{n_k}| + |x_{n_k} - y_{n_k}| \\ &\leq |x - x_{n_k}| + 1/n_k \rightarrow 0 \end{aligned}$$

as $k \rightarrow \infty$. By continuity of f at x , $f(x_{n_k}) \rightarrow f(x)$ and $f(y_{n_k}) \rightarrow f(x)$. But this contradicts $|f(x_n) - f(y_n)| \geq \varepsilon$ (with n_k in place of n). $\square$

Essentially the same argument gives the following strengthening of this theorem.

Theorem 2.2

Let $f : [a, b] \rightarrow \mathbb{R}$ where $-\infty < a < b < \infty$ be any function. Suppose there is a (finite, countable or uncountable) collection $\mathcal{C}$ of open intervals $I \subseteq \mathbb{R}$ such that if $F = [a, b] \setminus \bigcup_{I \in \mathcal{C}} I$, then f is continuous at every point in F . That is, the set of discontinuities $\mathcal{D}_f \subseteq \bigcup_{I \in \mathcal{C}} I$. Then,

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x - y| < \delta \text{ with } x \in F \text{ and } y \in [a, b] \text{ implies } |f(x) - f(y)| < \varepsilon.$$

This includes the statement that f is uniformly continuous on F , but the theorem says more. Note the hypothesis that f is continuous at every point in F is stronger than just saying that the restriction $f|_F$ is continuous on F . For example, taking $f : [0, 1] \rightarrow \mathbb{R}$ with $f(x) = \mathbb{1}_Q(x)$, f is nowhere continuous but $f|_{[0,1] \setminus Q} = 0$ is continuous.

Proof. Essentially the same as the proof of Theorem 2.1. The only additional point needed is that the limit point x belongs to F . This is true. If $x \in I$ for some $I \in \mathcal{C}$, since I is an open interval, this would mean that $x_{n_k} \in I$ for all sufficiently large k , but this contradicts the fact that $x_n \in F$ for all n . After this, the rest of the proof is the same. $\square$

Some revision on last year, let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function, say $m \leq f(x) \leq M$ for all $x \in [a, b]$ with $-\infty < m \leq M < \infty$. Let $P = \{a = a_0 < a_1 < \dots < a_n = b\}$ be a partition of $[a, b]$ and $I_j = [a_j, a_{j+1}]$. Define the upper sum of f with respect to P as

$$U(P, f) = \sum_{j=0}^{n-1} (a_{j+1} - a_j) \sup_{I_j} f$$

and the lower sum similarly as

$$L(P, f) = \sum_{j=0}^{n-1} (a_{j+1} - a_j) \inf_{I_j} f.$$

We know $m(b - a) \leq L(P, f) \leq U(P, f) \leq M(b - a)$. When we refine the partition by adding more points (a finite number), then the upper sum is non-increasing and the lower sum is non-decreasing. This is because for

$$a_j < a' < a_{j+1},$$

$$(a' - a_j) \sup_{[a_j, a']} f + (a_{j+1} - a') \sup_{[a', a_{j+1}]} f \leq (a_{j+1} - a_j) \sup_{[a_j, a_{j+1}]} f.$$

We then define the upper Riemann integral as $I^*(f) = \inf_P U(P, f)$ and the lower Riemann integral as $I_*(f) = \sup_P L(P, f)$. We know $m(b - a) \leq I_*(f) \leq I^*(f) \leq M(b - a)$. If $I_*(f) = I^*(f)$, we say that f is Riemann integrable on $[a, b]$ and write $\int_a^b f(x) dx = \int_a^b f$ for the common value, called the integral of f over $[a, b]$.

Theorem 2.3 – Riemann criterion for integrability

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is integrable if and only if for all $\varepsilon > 0$, there exists a partition P such that $U(P, f) - L(P, f) < \varepsilon$. That is,

$$\sum_{j=0}^{n-1} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) < \varepsilon.$$

Theorem 2.4

Let $f : [a, b] \rightarrow [A, B] \subseteq \mathbb{R}$ be integrable and $g : [A, B] \rightarrow \mathbb{R}$ be continuous. Then, $g \circ f : [a, b] \rightarrow \mathbb{R}$ is integrable.

Proof. First, by the uniform continuity of g , given $\varepsilon > 0$, there exists $\delta > 0$ such that $x, y \in [A, B]$ and $|x - y| < \delta$ implies $|g(x) - g(y)| < \varepsilon$. We also have, by the integrability of f , for any $\varepsilon' > 0$, a partition P of $[a, b]$ such that

$$U(P, f) - L(P, f) = \sum_{j=0}^{n-1} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) < \varepsilon'.$$

And we know

$$U(P, g \circ f) - L(P, g \circ f) = \sum_{j=0}^{n-1} (a_{j+1} - a_j) (\sup_{I_j} g \circ f - \inf_{I_j} g \circ f).$$

Let $J = \{j : \sup_{I_j} f - \inf_{I_j} f < \delta\}$, then

$$\sum_{j \in J} (a_{j+1} - a_j) (\sup_{I_j} g \circ f - \inf_{I_j} g \circ f) < \sum_{j \in J} (a_{j+1} - a_j) \varepsilon \leq (b - a) \varepsilon.$$

If $\sup_{I_j} f - \inf_{I_j} f \geq \delta$, then $\delta(a_{j+1} - a_j) \leq (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f)$, so

$$\delta \sum_{j \notin J} (a_{j+1} - a_j) \leq \sum_{j=0}^{n-1} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) < \varepsilon',$$

hence $\sum_{j \notin J} (a_{j+1} - a_j) < \varepsilon' / \delta$. We can choose $\varepsilon' = \delta \varepsilon$ then combine to get the result. $\square$

Corollary 2.1

If $g : [a, b] \rightarrow \mathbb{R}$ is continuous, then g is integrable.

Proof. We can check that $f(x) = x$ is integrable, and then apply Theorem 2.4. $\square$

Theorem 2.5 – Uniform limits of integrable functions are integrable

Let $f_n : [a, b] \rightarrow \mathbb{R}$ be bounded, Riemann integrable functions. Let $f : [a, b] \rightarrow \mathbb{R}$. If $f_n \rightarrow f$ uniformly on $[a, b]$, then f is bounded and integrable.

Proof.

$$\sup_{[a,b]} |f| \leq \sup_{[a,b]} |f - f_n| + \sup_{[a,b]} |f_n| \leq 1 + \sup_{[a,b]} |f_n|$$

if n is sufficiently large. Since $\sup_{[a,b]} |f_n| < \infty$, this shows that f is bounded.

Let $P = \{a = a_0 < a_1 < \cdots < a_m = b\}$ be any partition of $[a, b]$. Let $\varepsilon > 0$,

$$\begin{aligned} U(P, f) - L(P, f) &= \sum_{j=0}^{m-1} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) \\ &= \sum_{j=0}^{m-1} (a_{j+1} - a_j) (\sup_{I_j} (f - f_n + f_n) - \inf_{I_j} (f - f_n + f_n)) \\ &\leq \sum_{j=0}^{m-1} (a_{j+1} - a_j) (\sup_{I_j} (f - f_n) + \sup_{I_j} f_n - \inf_{I_j} (f - f_n) - \inf_{I_j} f_n) \\ &\leq U(P, f_n) - L(P, f_n) + 2(b - a) \sup_{[a,b]} |f - f_n|. \end{aligned}$$

Given $\varepsilon > 0$, we can first choose some N such that $2(b - a) \sup_{[a,b]} |f - f_N| < \varepsilon/2$ by uniform convergence. Then, we can choose a partition P_N such that $U(P_N, f_N) - L(P_N, f_N) < \varepsilon/2$ by the integrability of f_N . So, $U(P_N, f) - L(P_N, f) < \varepsilon$ and hence f is integrable. $\square$

The discussion below is non-examinable. Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and use $\mathcal{D}_f$ to denote the set of discontinuities of f . We know that there are functions with $\mathcal{D}_f \neq \emptyset$ which are integrable. For example, monotone functions with discontinuities are integrable, but $\mathcal{D}_f$ would have to be countable. How big a set $\mathcal{D}_f$ can we allow, and still maintain the integrability of f ?

Definition 2.2 – Null set

A subset $\mathcal{D} \subseteq \mathbb{R}$ is said to be a null set, or a set of Lebesgue measure zero, if for all $\varepsilon > 0$, there exists a countable (including possibly finite) collection of open intervals $I_j = (a_j, b_j)$ such that

$$\mathcal{D} \subseteq \bigcup_{j=1}^{\infty} I_j \text{ and } \sum_{j=1}^{\infty} |I_j| < \varepsilon$$

where $|I_j| = b_j - a_j$.

- (i) The empty set and any singleton $A = \{a\}$ are null sets.
- (ii) Any subset of a null set is null.
- (iii) Any countable union of null sets is null. Say E_j are null and let $E = \bigcup_{j=1}^{\infty} E_j$. Let $\varepsilon > 0$, for all j , there exist open intervals $I_j^{(k)}$ such that $E_j \subseteq \bigcup_{k=1}^{\infty} I_j^{(k)}$ and $\sum_{k=1}^{\infty} |I_j^{(k)}| < \varepsilon/2^j$. Then, $E \subseteq \bigcup_{j,k} I_j^{(k)}$ is the union of a countable collection of open intervals, and $\sum_{j,k=1}^{\infty} |I_j^{(k)}| < \sum_{j=1}^{\infty} \varepsilon/2^j = \varepsilon$.
- (iv) By (i) and (iii), any countable subset of $\mathbb{R}$ is a null set.
- (v) A null set need not be countable. For example, the standard Cantor set is a null set.
- (vi) Not every subset of $\mathbb{R}$ is a null set. Any open or closed interval is not null. This follows from the following lemma.

Lemma 2.1

If $(a, b) \subseteq \bigcup_{j=1}^{\infty} I_j$ where I_j are open intervals, then $b - a \leq \sum_{j=1}^{\infty} |I_j|$.

To prove this, take any $\tilde{a}, \tilde{b}$ with $a < \tilde{a} < \tilde{b} < b$. Then, there is a finite N such that $[\tilde{a}, \tilde{b}] \subset \bigcup_{j=1}^N I_j$ using Bolzano-Weierstrass. So, $\tilde{b} - \tilde{a} \leq \sum_{j=1}^N |I_j| \leq \sum_{n=1}^{\infty} |I_j|$ and we can let $\tilde{a} \rightarrow a$ and $\tilde{b} \rightarrow b$.

Theorem 2.6 – Lebesgue’s theorem on the Riemann integral

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then, f is Riemann integrable if and only if $\mathcal{D}_f$ is a null set.

Proof. From Analysis I, we saw that the integrability of f implies that $\mathcal{D}_f$ is null.

For the reverse direction, suppose that $\mathcal{D}_f$ is null. Let $\varepsilon > 0$, we have $\mathcal{D}_f \subseteq \bigcup_{j=1}^{\infty} I_j$ with $\sum_{j=1}^{\infty} |I_j| < \varepsilon$. Let $F = [a, b] \setminus \bigcup_{j=1}^{\infty} I_j$ and we have the following. Since f is continuous on F , there exists $\delta > 0$ such that for all $x \in F, y \in [a, b]$ with $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$. With this $\delta > 0$, take any partition $P = \{a = a_0 < a_1 < \dots < a_n = b\}$ with $a_{j+1} - a_j < \delta$.

$$\begin{aligned} U(P, f) - L(P, f) &= \sum_{j=0}^{n-1} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) \\ &= \sum_{j \in J} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) + \sum_{j \notin J} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) \end{aligned}$$

where $J = \{j : [a_j, a_{j+1}] \cap F \neq \emptyset\}$.

If $j \in J$, then there exists $x \in [a_j, a_{j+1}] \cap F$, so for any $y_1, y_2 \in [a_j, a_{j+1}]$, we have $f(y_1) - f(y_2) \leq |f(y_1) - f(x)| + |f(x) - f(y_2)| \leq \varepsilon + \varepsilon = 2\varepsilon$. So,

$$\sum_{j \in J} (a_{j+1} - a_j) (\sup_{I_j} f - \inf_{I_j} f) \leq 2\varepsilon(b - a).$$

If $j \notin J$, then $(a_j, a_{j+1}) \subseteq \bigcup_{k=1}^{\infty} I_k$, so $(a_j, a_{j+1}) \subseteq \bigcup_{k=1}^{\infty} I_k \cap (a_j, a_{j+1})$. And we can apply the lemma, $(a_{j+1} - a_j) \leq \sum_{k=1}^{\infty} |I_k \cap (a_j, a_{j+1})|$, so

$$\begin{aligned} \sum_{j \notin J} (a_{j+1} - a_j) &\leq \sum_{j \notin J} \sum_{k=1}^{\infty} |I_k \cap (a_j, a_{j+1})| \\ &= \sum_{k=1}^{\infty} \sum_{j \notin J} |I_k \cap (a_j, a_{j+1})| \\ &\leq \sum_{k=1}^{\infty} |I_k| < \varepsilon \end{aligned}$$

since (a_j, a_{j+1}) are disjoint. Combining everything, we get the result. $\square$

Many results on Riemann integration are very direct consequences of Lebesgue’s theorem. For example,

Corollary 2.2

If $f : [a, b] \rightarrow \mathbb{R}$ is monotone, then f is integrable.

Proof. The monotonicity of f implies that $\mathcal{D}_f$ is countable and hence null. $\square$

Corollary 2.3

If $f : [a, b] \rightarrow [A, B]$ is integrable and $g : [A, B] \rightarrow \mathbb{R}$ is continuous, then $g \circ f$ is integrable.

Proof. $g \circ f$ is bounded since g is bounded. The integrability of f implies that $\mathcal{D}_f$ is null. If f is continuous at some point x , then $g \circ f$ is also continuous at x since g is continuous. Hence, $\mathcal{D}_{g \circ f} \subseteq \mathcal{D}_f$ and $\mathcal{D}_{g \circ f}$ is null. $\square$

Corollary 2.4

If $f : [a, b] \rightarrow \mathbb{R}$ is integrable and $f_n \rightarrow f$ uniformly, then f is integrable.

Proof. $\mathcal{D}_{f_n}$ is null by Lebesgue's theorem. Hence, $\bigcup_{n=1}^{\infty} \mathcal{D}_{f_n}$ is null. But $\mathcal{D}_f \subseteq \bigcup_{n=1}^{\infty} \mathcal{D}_{f_n}$ since uniform convergence preserves points of continuity, so $\mathcal{D}_f$ is null. $\square$

Here's a new result that can be deduced from Lebesgue's theorem.

Corollary 2.5

If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, then $|f|$ is integrable on $[a, b]$. Moreover, $\int_a^b |f| = 0$ if and only if f is zero except on a null set, that is there exists a null set $N \subseteq [a, b]$ such that $f(x) = 0$ for all $x \in [a, b] \setminus N$. We say $f = 0$ almost everywhere on $[a, b]$.

Proof. As an exercise. $\square$

With this, we are back to examinable material.

3 Metric and normed spaces

Definition 3.1 – Metric space

Let X be any set. A metric, or a distance function, on X is a function $d : X \times X \rightarrow \mathbb{R}$ satisfying the following for any $x, y, z \in X$.

- (i) $d(x, y) \geq 0$ with $d(x, y) = 0$ if and only if $x = y$.
- (ii) $d(x, y) = d(y, x)$.
- (iii) $d(x, y) \leq d(x, z) + d(z, y)$.

We call such a pair (X, d) a metric space.

Definition 3.2 – Normed space

Let V be a real vector space. A norm on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ satisfying the following for any $x, y \in V$ and $\lambda \in \mathbb{R}$.

- (i) $\|x\| \geq 0$ with $\|x\| = 0$ if and only if $x = 0$.
- (ii) $\|\lambda x\| = |\lambda| \|x\|$.
- (iii) $\|x + y\| \leq \|x\| + \|y\|$.

We call such a pair $(V, \|\cdot\|)$ a normed vector space or normed space.

Proposition 3.1

If $(V, \|\cdot\|)$ is a normed space, and if $d : V \times V \rightarrow \mathbb{R}$ is defined by $d(x, y) = \|x - y\|$, then (V, d) is a metric space.

Proof. An easy exercise. □

Example 3.1 – Finite-dimensional normed spaces

Consider the Euclidean space $\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}$ with $n \in \mathbb{N}$ fixed. Taking $V = \mathbb{R}^n$ with its usual vector space structure, we can define several useful norms on V . Write $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

- (i) The Euclidean norm, or the l_2 norm, defined by $\|x\|_2 = (\sum_{i=1}^n x_i^2)^{1/2}$. It is easy to check the requirements of a norm, expect perhaps the triangle inequality. By the Cauchy-Schwarz inequality,

$$\|x + y\|_2^2 = \sum_{i=1}^n (x_i + y_i)^2 = \|x\|_2^2 + \|y\|_2^2 + 2 \sum_{i=1}^n x_i y_i \leq \|x\|_2^2 + \|y\|_2^2 + \|x\|_2 \|y\|_2.$$

- (ii) The l_1 norm, defined by $\|x\|_1 = \sum_{i=1}^n |x_i|$.

- (iii) The l_∞ norm, defined by $\|x\|_\infty = \sup \{|x_i| : 1 \leq i \leq n\}$.

More generally, we can define

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

It turns out that for $p \geq 1$, this is a norm. Moreover, $\|x\|_p \rightarrow \|x\|_\infty$ as $p \rightarrow \infty$.

Example 3.2 – Spaces of sequences of real numbers

Now, let's look at infinite-dimensional normed spaces. Let $\mathbb{R}^\mathbb{N} = \{\text{sequences } (x_k)_{k \in \mathbb{N}} \text{ of real numbers}\}$. This is a vector space with $(x_k) + (y_k) = (x_k + y_k)$ and $\lambda(x_k) = (\lambda x_k)$. Write $x = (x_k) \in \mathbb{R}^\mathbb{N}$.

- (i) The space $l_1 = \{(x_k) \in \mathbb{R}^\mathbb{N} : \sum_{k=1}^\infty |x_k| < \infty\}$. This is a linear subspace of $\mathbb{R}^\mathbb{N}$. We can form this into a normed space by defining $\|x\|_1 = \sum_{k=1}^\infty |x_k|$.
- (ii) The space $l_2 = \{(x_k) \in \mathbb{R}^\mathbb{N} : \sum_{k=1}^\infty x_k^2 < \infty\}$ is a linear subspace. We can define $\|x\|_2 = (\sum_{k=1}^\infty |x_k|^2)^{1/2}$.
- (iii) The space $l_\infty = \{(x_k) \in \mathbb{R}^\mathbb{N} : \sup_{k \geq 1} |x_k| < \infty\}$, which is the space of bounded sequences. Define $\|x\|_\infty = \sup_{k \geq 1} |x_k|$.

It is easy to check that these are all normed spaces. More generally, $l_p = \{(x_k) \in \mathbb{R}^\mathbb{N} : \sum_{k=1}^\infty |x_k|^p < \infty\}$ is a subspace of $\mathbb{R}^\mathbb{N}$ and we can define $\|x\|_p = (\sum_{k=1}^\infty |x_k|^p)^{1/p}$. Note that $\|(x_k)\|_p = \lim_{n \rightarrow \infty} \|(x_1, x_2, \dots, x_n)\|_p$ and hence the triangle inequality for l_p is a consequence of the triangle inequality for the finite dimensional l_p norm on $\mathbb{R}^n$.

Example 3.3 – The space of continuous function on a bounded, closed interval

Consider $V = C([a, b]) = \{f : [a, b] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$. This is a vector space under pointwise addition and scalar multiplication of functions.

- (i) The L^1 norm is $\|f\|_1 = \int_a^b |f(x)| dx$.
- (ii) The L^2 norm is $\|f\|_2 = (\int_a^b |f(x)|^2 dx)^{1/2}$.
- (iii) The L^∞ norm, or the uniform norm, is $\|f\|_\infty = \sup_{x \in [a, b]} |f(x)|$.

It is easy to check that these are all norms. The triangle inequality for L^2 requires the Cauchy-Schwarz inequality for integrals

$$\int_a^b |fg| dx \leq \sqrt{\int_a^b |f|^2 dx \int_a^b |g|^2 dx}.$$

Proof. Let $\varphi(t) = \int_a^b (|f| - t|g|)^2 dx \geq 0$ for all $t \in \mathbb{R}$. This is a quadratic in t which is non-negative for all t . So, its discriminant is non-positive which gives the result. $\square$

By Proposition 3.1, all of these examples are naturally metric spaces as well. For example, the Euclidean metric $d(x, y) = \|x - y\|_2$ on $\mathbb{R}^n$ and the uniform metric $d(f, g) = \|f - g\|_\infty = \sup_{x \in [a, b]} |f(x) - g(x)|$ on $C([a, b])$.

Integral norms, such as L^1 , are more naturally defined on the larger space of integrable functions $\mathcal{R}([a, b])$ which is a vector space under pointwise addition and scalar multiplication, and $C([a, b]) \subseteq \mathcal{R}([a, b])$. The slight problem is that for $f \in \mathcal{R}([a, b])$, $\|f\|_1$ does not imply $f = 0$ on $[a, b]$. But by Corollary 2.5, we know that $\|f\|_1$ implies $f = 0$ almost everywhere on $[a, b]$. We can then define an equivalence relation $\sim$ on $\mathcal{R}([a, b])$ by setting $f \sim g$ if $f = g$ almost everywhere on $[a, b]$. On $\mathcal{R}([a, b]) / \sim$, we have $[f] + [g] = [f + g]$ and $\lambda[f] = [\lambda f]$ being well-defined, that is independent of the representative. Define $\|\cdot\|_1$ on $\mathcal{R}([a, b]) / \sim$ by $\|[f]\|_1 = \int_a^b |f|$, and $(\mathcal{R}([a, b]) / \sim, \|\cdot\|_1)$ is a normed space.

Example 3.4 – Discrete metric

For any set X ,

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

is the discrete metric.

Example 3.5 – New metrics from given ones

Let (X, d) be a metric space and define $g : X \times X \rightarrow \mathbb{R}$ by $g(x, y) = \min\{1, d(x, y)\}$. Then, g is a metric. Or, define $h : X \times X \rightarrow \mathbb{R}$ by $h(x, y) = d(x, y) / (1 + d(x, y))$. Again, this is a metric. These are both bounded metrics, for all $x, y \in X$, we have $g(x, y) \leq 1$ and $h(x, y) < 1$. If $X = \mathbb{R}^2$,

$$d(x, y) = \begin{cases} \|x - y\|_2 & \text{if } x = ty \text{ for some } t \in \mathbb{R} \\ \|x\|_2 + \|y\|_2 & \text{otherwise} \end{cases}$$

is a metric, called the “French railway” metric.

Definition 3.3 – Metric subspace

Let (X, d) be a metric space and $Y \subseteq X$ be any subset. Then, the restriction $d|_{Y \times Y} : Y \times Y \rightarrow \mathbb{R}$ is a metric on Y , called the induced metric or the subspace metric.

Definition 3.4 – Open and closed sets

Let (X, d) be a metric space.

- (i) For any $a \in X$ and any $r > 0$, the open ball with radius r and centre a is the set $B_r(a) = \{x \in X : d(x, a) < r\}$.
- (ii) A subset $U \subseteq X$ is open if for all $a \in U$, there exists $r > 0$ such that $B_r(a) \subseteq U$.
- (iii) A subset $E \subseteq X$ is closed if $X \setminus E$ is open.

Note that these balls are not necessarily "circular". It can be all sorts of shapes depending on the metric. Also, being open or closed is not mutually exclusive. Moreover, the property of being open or closed for a subset is relative to the containing space.

Example 3.6

Consider $X = \mathbb{R}$ with the Euclidean metric and $Y = [0, 1) \cup \{2\}$ with the induced metric. $[0, 1)$ is neither open nor closed in X and $\{2\}$ is closed but not open in X . However, $[0, 1)$ and $\{2\}$ are both open and closed in Y .

Proposition 3.2

Let (X, d) be a metric space.

- (i) Any open ball $B_r(a)$ is an open set.
- (ii) Any singleton $\{x\}$ with $x \in X$ is closed.

Note that open balls can be closed sets, and singletons can be open sets. For example, $X = \mathbb{R}^2$ with the discrete metric, $B_r(a) = \{a\}$ for any $r < 1$.

Proof.

- (i) Let $y \in B_r(a)$ and $r_1 = r - d(y, a)$. Then, $r_1 > 0$ since $d(y, a) < r$. Let $z \in B_{r_1}(y)$, $d(z, a) \leq d(z, y) + d(y, a) < r_1 + d(y, a) = r$ so $z \in B_r(a)$. Hence, $B_{r_1}(y) \subseteq B_r(a)$ and $B_r(a)$ is open.
- (ii) If $z \in X \setminus \{x\}$, then $r = d(z, x) > 0$. So, $x \in B_r(z)$ and so $B_r(z) \subseteq X \setminus \{x\}$. Hence, $X \setminus \{x\}$ is open and $\{x\}$ is closed.

□

Theorem 3.1

Let (X, d) be a metric space.

- (i) The union of any (possibly uncountable) collection of open sets is open.
- (ii) The intersection of any finite collection of open sets is open.
- (iii) The empty set $\emptyset$ and X are both open.

By taking complements, we get

Theorem 3.2

- (i) The intersection of any (possibly uncountable) collection of closed sets is closed.
- (ii) The union of any finite collection of closed sets is closed.
- (iii) The empty set $\emptyset$ and X are both closed.

The collection being finite is important for (ii). Take $X = \mathbb{R}$, an infinite intersection of open sets $\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$ is not open. An infinite union of closed sets $\bigcup_{n=1}^{\infty} (\mathbb{R} \setminus (-1/n, 1/n)) = \mathbb{R} \setminus \{0\}$ is not closed.

Proof. As an exercise.

□

Definition 3.5 – Convergence in a metric space

Let (X, d) be a metric space. A sequence (x_k) in X is said to converge to a point $x \in X$ if

$$d(x_k, x) \rightarrow 0$$

as real numbers. That is, for all $\varepsilon > 0$, there exists N such that $k \geq N$ implies $d(x_k, x) < \varepsilon$. Or, for all $\varepsilon > 0$, there exists N such that $k \geq N$ implies $x_k \in B_\varepsilon(x)$.

It is clear for the ε - N definition that if $x_k \rightarrow x$, then $x_{k_j} \rightarrow x$ for any subsequence $(x_{k_j})_j$.

Proposition 3.3

If (X, d) is a metric space and (x_k) is a sequence in X such that $x_k \rightarrow x$ and $x_k \rightarrow y$, then $x = y$.

Proof. $d(x, y) \leq d(x, x_k) + d(x_k, y)$. As $k \rightarrow \infty$, $d(x, y) = 0$ so $x = y$. □

It is possible that the same sequence in X has different limits with respect to different metrics in X . For example, taking $X = \mathbb{R}$, let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x = 1 \\ x & \text{otherwise} \end{cases}$$

and $d_1(x, y) = |f(x) - f(y)|$. Then, the sequence $x_k = 1/k \rightarrow 0$ in the Euclidean metric but $x_k \rightarrow 1$ in d_1 . This can happen even in normed spaces, with respect to two different norms. If the normed space is finite dimensional, this cannot happen because, as we shall see later, any two norms on a finite-dimensional vector space are equivalent in a certain precise sense.

Proposition 3.4

Convergence in $\mathbb{R}^n$ with respect to the Euclidean norm is equivalent to the convergence of the coordinates as real numbers. Let $x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)})$ be a sequence in $\mathbb{R}^n$ and $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then

$$x^{(k)} \rightarrow x \text{ in } \|\cdot\|_2 \iff x_j^{(k)} \rightarrow x_j \text{ for all } j \in \{1, \dots, n\}.$$

Proof. For the $\implies$ direction, fix $\varepsilon > 0$, there exists N such that $k \geq N$ implies $\|x^{(k)} - x\|_2 < \varepsilon$. That is, $\sum_{j=1}^n (x_j^{(k)} - x_j)^2 < \varepsilon^2$ so $|x_j^{(k)} - x_j| < \varepsilon$ for all j . Hence, $x_j^{(k)} \rightarrow x_j$ for all j .

For the $\impliedby$ direction, for all $j \in \{1, \dots, n\}$, there exists N_j such that $|x_j^{(k)} - x_j| < \varepsilon/\sqrt{n}$ for $k \geq N_j$. Let $N = \max\{N_1, \dots, N_n\}$, then $k \geq N$ implies $\sum_{j=1}^n |x_j^{(k)} - x_j|^2 < \varepsilon^2$. That is, $\|x^{(k)} - x\|_2 < \varepsilon$ so $x^{(k)} \rightarrow x$ in $\|\cdot\|_2$. □

By the equivalence of norms in finite-dimensional spaces, this holds with respect to any norm in $\mathbb{R}^n$. The convergence in $(C([a, b]), \|\cdot\|_\infty)$ is just uniform convergence as we have seen earlier.

Definition 3.6 – Bounded set

Let (X, d) be a metric space. A subset $E \subseteq X$ is bounded if $E \subseteq B_R(a)$ for some $a \in X$ and some $R > 0$.

Theorem 3.3 – Bolzano-Weierstrass in $\mathbb{R}^n$

Every bounded sequence in $\mathbb{R}^n$ (say, with respect to the Euclidean metric) has a convergent subsequence.

Proof. We proceed by induction on n . The case $n = 1$ is Bolzano-Weierstrass on the real line, which was proved in Analysis I. Let $n \geq 2$ and assume by induction that the theorem holds in $\mathbb{R}^{n-1}$. Let $(x^{(k)})$ be a bounded sequence in $\mathbb{R}^n$, so $\|x^{(k)}\|_2^2 \leq R^2$ for some R and all k . Write

$$x^{(k)} = (\underbrace{x_1^{(k)}, x_2^{(k)}, \dots, x_{n-1}^{(k)}}_{\text{call this } y^{(k)} \in \mathbb{R}^{n-1}}, x_n^{(k)})$$

so $\|y^{(k)}\|_2^2 + |x_n^{(k)}|^2 \leq R^2$ and $(y^{(k)})$ is a bounded sequence in $\mathbb{R}^{n-1}$. By the induction hypothesis, there exists a subsequence $(k_j)_j$ and a point $y \in \mathbb{R}^{n-1}$ such that $y^{(k_j)} \rightarrow y$ in $\mathbb{R}^{n-1}$. Since $(x_n^{(k_j)})_j$ is a bounded in $\mathbb{R}$, we have a further subsequence $(x_n^{(k_{j_l})})_l$ and a point $x_n \in \mathbb{R}$ such that $x_n^{(k_{j_l})} \rightarrow x_n$. Let $x = (y, x_n) \in \mathbb{R}^n$. By Proposition 3.4, $x^{(k_{j_l})} \rightarrow x$ in $\mathbb{R}^n$ with respect to $\|\cdot\|_2$. $\square$

Example 3.7

This does not work with infinite dimensions. Consider $(l_\infty, \|\cdot\|_\infty)$ and $e^{(k)} = (0, 0, \dots, 0, 1, 0, \dots)$ where the one is in the k -th position. $\|e^{(k)}\|_\infty = 1$ so it is a bounded sequence in l_∞ . For all fixed j , $e_j^{(k)} \rightarrow 0$ as $k \rightarrow \infty$. $(e^{(k)})$ does not have a convergent subsequence. If it did, say $\|e^{(k_l)} - e\|_\infty \rightarrow 0$ as $l \rightarrow \infty$, then $|e_j^{(k_l)} - e_j| \leq \|e^{(k_l)} - e\|_\infty \rightarrow 0$ as $l \rightarrow \infty$. Since $e_j^{(k_l)} \rightarrow 0$ as $l \rightarrow \infty$ for all fixed j , we must have that $e = 0 \in l_\infty$. But $\|e^{(k_l)} - e\|_\infty = \|e^{(k_l)}\|_\infty = 1$ which contradicts $\|e^{(k_l)} - e\|_\infty \rightarrow 0$.

In fact, for normed spaces $(V, \|\cdot\|)$, the Bolzano-Weierstrass property (every bounded sequence has a convergent subsequence) is equivalent to the finite-dimensionality of V .

Definition 3.7 – Limit points, isolated points, closure and interior

Let (X, d) be a metric space and $E \subseteq X$ be any subset. Then,

- (i) a point $x \in X$ is a limit point of E if there is a sequence (x_k) in E with $x_k \neq x$ for all k and $x_k \rightarrow x$;
- (ii) a point $x \in X$ is an isolated point of E if $x \in E$ and x is not a limit point of E . Or equivalently, if there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \cap E = \{x\}$.
- (iii) the closure of E is the set $\bar{E} = \{x \in X : x = \lim_{k \rightarrow \infty} x_k \text{ for some sequence } (x_k) \text{ in } E\} = E \cup \{\text{limit points of } E\}$;
- (iv) a point $x \in X$ is an interior point of E if there exists $r > 0$ such that $B_r(x) \subseteq E$;
- (v) the interior of E is the set $\mathring{E} = \{x \in X : x \text{ is an interior point of } E\}$.

Example 3.8

Take $X = \mathbb{R}$ with the Euclidean metric and $E = [0, 1) \cup \{2\}$. The limit points of E are $[0, 1]$. 2 is an isolated point. $\bar{E} = [0, 1] \cup \{2\}$ and $\mathring{E} = (0, 1)$.

Proposition 3.5

For any $E \subseteq X$, $\bar{E}$ is a closed set. In fact,

$$\bar{E} = \bigcap_{\substack{F \text{ is closed in } X \\ E \subseteq F}} F,$$

that is $\bar{E}$ is the smallest closed set in X containing E .

Proof. On example sheet 2. □

Proposition 3.6

Let $E \subseteq X$, then the following are equivalent:

- (i) If (x_k) is a sequence in E with $x_k \rightarrow x \in X$, then $x \in E$.
- (ii) $E = \bar{E}$.
- (iii) E is closed in X .

Proof. Directly follows from the definitions of closure and closed set, and from Proposition 3.5. □

Definition 3.8 – Cauchy sequence

Let (X, d) be a metric space. A sequence (x_k) in X is a Cauchy sequence if

$$\forall \varepsilon > 0, \exists N \text{ such that } k, l \geq N \text{ implies } d(x_k, x_l) < \varepsilon.$$

Proposition 3.7

- (i) If (x_k) is a convergent sequence in X , then (x_k) is Cauchy.
- (ii) Any Cauchy sequence is bounded.
- (iii) If (x_k) is a Cauchy sequence with a convergent subsequence $x_{k_l} \rightarrow x$, then $x_k \rightarrow x$.

Proof.

- (i) If $x_k \rightarrow x$, then $d(x_k, x_l) \leq d(x_k, x) + d(x, x_l) \rightarrow 0$ so it is Cauchy.
- (ii) If (x_k) is Cauchy, then there exists N such that $d(x_k, x_N) < 1$ for all $k \geq N$. Taking $R = \max\{1, d(x_1, x_N), d(x_2, x_N), \dots, d(x_{N-1}, x_N)\}$, then $x_k \in B_{R+1}(x_N)$ for all k .
- (iii) For all $\varepsilon > 0$, there exists N such that $d(x_n, x_m) < \varepsilon$ for all $n, m \geq N$. If $l \geq N$, then since $k_l \geq l$, $d(x_{k_l}, x_l) < \varepsilon$. So, $d(x_l, x) \leq d(x_{k_l}, x_l) + d(x_{k_l}, x) < 2\varepsilon$ for sufficiently large l .

□

Definition 3.9 – Completeness

A metric space (X, d) is complete if every Cauchy sequence in X converges to some element in X . A normed space $(V, \|\cdot\|)$ is complete if V with the metric defined by $\|\cdot\|$ is complete.

Theorem 3.4

$\mathbb{R}^n$ (with the Euclidean metric) is complete.

Proof. If $(x^{(k)})$ is a Cauchy sequence in $\mathbb{R}^n$, then by applying the definition of a Cauchy sequence, $(x_j^{(k)})_k$ is Cauchy in $\mathbb{R}$ for each $j \in \{1, \dots, n\}$. So, $x_j^{(k)} \rightarrow x_j$ for some $x_j \in \mathbb{R}$. Hence, $x^{(k)} \rightarrow x = (x_1, \dots, x_n)$. $\square$

This theorem implies that any finite-dimensional normed space is complete.

Theorem 3.5

$(C([a, b]), \|\cdot\|_\infty)$ is complete.

Proof. Since uniformly Cauchy sequences converge (Theorem 1.3) and uniform limits of continuous functions are continuous (Corollary 1.1). $\square$

Theorem 3.6

$(l_\infty, \|\cdot\|_\infty)$, $(l_1, \|\cdot\|_1)$ and $(l_2, \|\cdot\|_2)$ are all complete normed spaces.

Proof. l_1 and l_2 are left as exercises. Sequences are just functions $\mathbb{N} \rightarrow \mathbb{R}$, so l_∞ is the set of bounded functions $\mathbb{N} \rightarrow \mathbb{R}$. Any Cauchy sequence in l_∞ then is a uniformly Cauchy sequence of bounded functions $\mathbb{N} \rightarrow \mathbb{R}$. Since uniformly Cauchy sequences converge uniformly, all that is left to check is that the limit is in the space l_∞ . This follows by the fact that if $f_n : E \rightarrow \mathbb{R}$ is a sequence of bounded functions, and if (f_n) converges uniformly, then the limit is bounded. $\square$

When does a subset of a complete metric space remain complete as a metric space with the induced metric?

Theorem 3.7

Let (X, d) be a complete metric space and $Y \subseteq X$ be any subset. Y with respect to the induced metric is complete if and only if Y is closed in X .

Proof. Suppose $(Y, d|_{Y \times Y})$ is complete. Let (x_k) be a sequence in Y with $x_k \rightarrow x \in X$, so (x_k) is Cauchy in X . Since the metric on Y is the induced metric, this means that (x_k) is Cauchy in Y , so (x_k) converges to some element $z \in Y$. By the uniqueness of limits, $z = x$ so $x \in Y$. Hence, Y is closed.

Conversely, suppose Y is closed. Let (x_k) be Cauchy in Y . Since X is complete and the metric on Y is the induced metric, there exists $x \in X$ such that $x_k \rightarrow x$. Since Y is closed, $x \in Y$. Hence, $(Y, d|_{Y \times Y})$ is complete. $\square$

Example 3.9

Take $X = (0, 1)$ with the Euclidean metric. X is not complete since $(1/n)_{n \geq 1}$ is Cauchy but has no limit in X . Or, $X \subseteq \mathbb{R}$ and $\mathbb{R}$ is complete. But X is not closed in $\mathbb{R}$, so by Theorem 3.7, X is not complete.

Example 3.10

Take $X = \mathbb{R} \setminus \{0\}$ with the Euclidean metric and $Y = (0, \infty)$. Y is closed in X , but is not complete, since X is not complete either so Theorem 3.7 does not apply. $Y_1 = [1, \infty)$ is complete and is closed.

Example 3.11 – An incomplete normed space

Let $l_0 = \{(x_k) \in \mathbb{R}^{\mathbb{N}} : x_k = 0 \text{ for all but finitely many } k\}$. l_0 is a subspace of l_∞ and take $\|\cdot\|_\infty$ to be the norm on l_0 . Let $(x^{(n)}) = (1, 1/2, 1/3, \dots, 1/n, 0, 0, \dots)$. This converges in l_∞ to $x = (1, 1/2, 1/3, \dots)$ where $x_i = 1/i$ for all i since $\|x^{(n)} - x\|_\infty = 1/(n+1) \rightarrow 0$ as $n \rightarrow \infty$. But $x \notin l_0$ so l_0 is not closed, and hence not complete by Theorem 3.7.

Example 3.12

Let $X = \{f \in C([a, b]) : \|f(x)\|_\infty \leq R\}$. X is closed in $C([a, b])$, and $C([a, b])$ is complete. So, (X, d_∞) is complete by the theorem, where d_∞ is the metric defined on X by $\|\cdot\|_\infty$.

Example 3.13 – Incompleteness of the L^1 norm with respect to Riemann integration

$(C([a, b]), L^1)$ is incomplete from example sheet 2. Let $\tilde{\mathcal{R}}([a, b]) = \mathcal{R}([a, b]) / \sim$ where $f \sim g$ means $f = g$ except on a null set as before. But, $(\tilde{\mathcal{R}}([a, b]), L^1)$ is not complete. This is the real drawback of Riemann's theory of integration, and is very successfully fixed by Lebesgue integration.

We are working towards the following theorem:

If V is a finite-dimensional (real) vector space, and if $\|\cdot\|$ and $\|\cdot\|'$ are two norms on V , then $\|\cdot\|$ and $\|\cdot\|'$ are Lipschitz equivalent. That is, there exists constants $c, C > 0$ such that $c\|x\|' \leq \|x\| \leq C\|x\|'$ for all $x \in V$.

For this theorem and for many other purposes in analysis, we need the notion of

Definition 3.10 – Sequential compactness

- (i) A metric space (X, d) is sequentially compact if every sequence (x_k) in X has a subsequence (x_{k_j}) converging to an element in X .
- (ii) A subset $K \subseteq X$ is sequentially compact if $(K, d|_{K \times K})$ is sequentially compact. Equivalently, every sequence in K has a subsequence that converges to an element in K .

There is another notion called compactness that applies to topological spaces. Metric spaces are an example of topological spaces, and it is a theorem a metric space is sequentially compact if and only if it is compact. For this reason, we will just write compact to mean sequentially compact.

Example 3.14

- (i) $I = [0, 1]$ is a compact subset of $\mathbb{R}$. To prove this, use Bolzano-Weierstrass and the fact that I is closed in $\mathbb{R}$.
- (ii) $I = (0, 1) \subseteq \mathbb{R}$ is not compact. $(1/n)_{n \geq 1}$ has no convergent subsequence in $(0, 1)$.
- (iii) $\mathbb{R}$ is not compact. $(n)_{n \geq 1}$ has no convergent subsequence.
- (iv) $S = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$ is compact. By Bolzano-Weierstrass in the ambient space $\mathbb{R}^n$, there exists a subsequence (x_{k_j}) and $x \in \mathbb{R}^n$ such that $\|x_{k_j} - x\|_2 \rightarrow 0$. Since $|\|x_{k_j}\|_2 - \|x\|_2| \leq \|x_{k_j} - x\|_2$ and $\|x_{k_j}\|_2 = 1$ for all j , $\|x\| = 1$ and $x \in S$.

Theorem 3.8

If a metric space (X, d) is compact, then it is complete and bounded.

Proof. For completeness, let (x_k) be a Cauchy sequence. By compactness, (x_k) has a convergent subsequence (x_{k_j}) . Since (x_k) is Cauchy, this means that the whole sequence (x_k) is convergent. So, X is complete.

For boundedness, fix $x_0 \in X$. If X is not bounded, then for all $k \in \mathbb{N}$, there exists x_k with $d(x_k, x_0) \geq k$. By compactness, (x_k) has a subsequence $(x_{k_j}) \rightarrow x \in X$. But this is a contradiction since $k_j \leq d(x_{k_j}, x_0) \leq d(x_{k_j}, x) + d(x, x_0) \leq 1 + d(x, x_0)$. $\square$

Theorem 3.9

If $K \subseteq X$ is a compact subset of a metric space X , then K is closed and bounded.

Proof. If $(x_k) \subseteq K$ is such that $x_k \rightarrow x \in X$, then by compactness of K , we get a subsequence (x_{k_j}) and $y \in K$ such that $x_{k_j} \rightarrow y$. So $x = y$, hence $x \in K$. So, K contains all its limit points, and hence K is closed. K is bounded by Theorem 3.8. $\square$

The converse of Theorem 3.8 and the converse of Theorem 3.9 are both false.

Example 3.15

The closed unit ball $D = \overline{B_1(0)}$ in l_∞ is not compact since $e^{(k)} = (0, 0, \dots, 0, 1, 0, \dots)$ where the one is in the k -th position has no convergent subsequence. D is closed, bounded and is complete (since l_∞ is complete) with the induced metric.

Example 3.16

Take $X = \mathbb{R}$ with $d(x, y) = \min\{1, |x - y|\}$. B is bounded since $X \subseteq B_2(0)$. (X, d) is complete. However, it is not compact since $(n)_{n \geq 1}$ has no convergent subsequence, $d(x_n, x) = 1$ for any $x \in X$ and sufficiently large n .

Theorem 3.10

$K \subseteq \mathbb{R}^n$ is compact $\iff K$ is closed and bounded.

Proof. Use Theorem 3.9 for the $\implies$ direction.

For the $\impliedby$ direction, let $(x_k) \subseteq K$. Since K is bounded, by Bolzano-Weierstrass in $\mathbb{R}^n$, there is a subsequence (x_{k_j}) and $x \in \mathbb{R}^n$ with $x_{k_j} \rightarrow x$. Since K is closed, $x \in K$, so K is compact. $\square$

For general metric spaces, completeness together with a certain stronger form of boundedness (called total boundedness) guarantees compactness.

Definition 3.11 – Total boundedness

A metric space (X, d) is totally bounded if for all $\varepsilon > 0$, there exists a finite collection of points $x_1, \dots, x_N \in X$ such that

$$X \subseteq \bigcup_{i=1}^N B_\varepsilon(x_i).$$

Theorem 3.11

A metric space (X, d) is compact $\iff X$ is complete and totally bounded.

Proof. On example sheet 3. □

Definition 3.12 – Continuous maps

Let (X, d) and (X', d') be metric spaces, and let $f : X \rightarrow X'$ be a map. We say f is continuous at a point $x \in X$ if

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon, x) > 0 \text{ such that } d(y, x) < \delta \implies d'(f(y), f(x)) < \varepsilon.$$

Or equivalently,

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x))).$$

We say f is continuous if f is continuous at every $x \in X$.

Theorem 3.12

Let $f : X \rightarrow X'$. Then, f is continuous at $x \in X$ if and only if for every sequence $(x_k) \subseteq X$ with $x_k \rightarrow x$, we have that $f(x_k) \rightarrow f(x)$ in X' .

Proof. Suppose that f is continuous at x . Let $(x_k) \subseteq X$ with $x_k \rightarrow x$. Let $\varepsilon > 0$, then there exists $\delta > 0$ such that $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$. Since $x_k \in B_\delta(x)$ for all sufficiently large k , $x_k \in f^{-1}(B_\varepsilon(f(x)))$, that is $d'(f(x_k), f(x)) < \varepsilon$.

For the other direction, if f is not continuous at x , then there exists $\varepsilon > 0$ such that $B_{1/k}(x) \not\subseteq f^{-1}(B_\varepsilon(f(x)))$ for all $k \in \mathbb{N}$. So pick any $x_k \in B_{1/k}(x) \setminus f^{-1}(B_\varepsilon(f(x)))$, then we have a sequence (x_k) such that $x_k \rightarrow x$, but $f(x_k) \not\rightarrow f(x)$. □

Theorem 3.13

Let (X, d) and (X', d') be metric spaces and $f : X \rightarrow X'$, then the following are equivalent:

- (i) f is continuous;
- (ii) for every sequence $x_k \rightarrow x$, we have $f(x_k) \rightarrow f(x)$;
- (iii) $f^{-1}(V) = \{x \in X : f(x) \in V\}$ is open in X for every open $V \subseteq X'$;
- (iv) $f^{-1}(E)$ is closed in X for every closed $E \subseteq X'$.

Proof. As an exercise. □

Definition 3.13 – Uniformly continuous maps

Let (X, d) and (X', d') be metric spaces. $f : X \rightarrow X'$ is uniformly continuous if

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0 \text{ such that } x, y \in X \text{ and } d(x, y) < \delta \implies d'(f(x), f(y)) < \varepsilon.$$

Or equivalently,

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x))) \text{ for all } x \in X.$$

Definition 3.14 – Lipschitz maps

We say that $f : X \rightarrow X'$ is Lipschitz if there exists a constant L such that $d'(f(x), f(y)) \leq L d(x, y)$ for all $x, y \in X$.

We can define the Lipschitz constant L as

$$\text{Lip}(f) = \sup_{x \neq y} \frac{d'(f(x), f(y))}{d(x, y)}.$$

Clearly, f being Lipschitz on X implies f is uniformly continuous on X (by taking $\delta = \varepsilon/L$) implies f is continuous on X . But, f being continuous does not imply that f is uniformly continuous. For example, $f : (0,1) \rightarrow \mathbb{R}$ with $f(x) = 1/x$. Also, f being uniformly continuous does not imply that f is Lipschitz. For example, $f : [0,1] \rightarrow \mathbb{R}$ with $f(x) = x^{1/2}$. f is uniformly continuous since it is continuous on a closed, bounded interval. f is not Lipschitz since $|f(x) - f(0)|/(x - 0) \rightarrow x^{-1/2} \rightarrow \infty$ as $x \rightarrow 0$.

Theorem 3.14

Let (X, d) be a compact metric space and (X', d') be any metric space. Let $f : X \rightarrow X'$ be continuous. Then,

- (i) f is uniformly continuous.
- (ii) $f(X) = \{f(x) : x \in X\}$ is compact in X' . That is, the continuous image of a compact space is compact.
- (iii) $f(X)$ is closed and bounded in X' .
- (iv) if $X' = \mathbb{R}$ with the Euclidean metric, then there exists $x^*, x_* \in X$ such that $f(x^*) = \sup f(X)$ and $f(x_*) = \inf f(X)$.

Proof.

- (i) We argue exactly as in the proof for uniform continuity of a continuous function on a closed, bounded interval.
- (ii) Let (y_k) be a sequence in $f(X)$. So, $y_k = f(x_k)$ for some $x_k \in X$. By the completeness of X , there exist a subsequence (x_{k_j}) and $x \in X$ such that $x_{k_j} \rightarrow x$ as $j \rightarrow \infty$. By continuity of f at x , $f(x_{k_j}) \rightarrow f(x) \in f(X)$. Letting $y = f(x)$, we have found the subsequence $y_{k_j} = f(x_{k_j})$ such that $y_{k_j} \rightarrow y \in f(X)$. Hence, $f(X)$ is compact.
- (iii) Follows from (ii) and Theorem 3.10.
- (iv) For a bounded subset $E \subseteq \mathbb{R}$, we have either $\sup E$ is in E or $\sup E$ is a limit point of E . Taking $E = f(X)$, we have that E is also closed, so $\sup E$ is in E . That is, there exists $x^* \in X$ such that $f(x^*) = \sup f(X)$. Similar reasoning applies to $\inf f(X)$.

□

Definition 3.15 – Topology

Let (X, d) be a metric space. The topology on X induced by d is the collection of open subsets of X .

Definition 3.16 – Topologically/Lipschitz equivalent metrics

- (i) Two metrics d and d' on X are topologically equivalent if they induce the same topology on X . That is, for any $U \subseteq X$, U is open with respect to d if and only if U is open with respect to d' .
 - (ii) d and d' are Lipschitz equivalent if there are fixed constants $a, b > 0$ such that $a d(x, y) \leq d'(x, y) \leq b d(x, y)$ for all $x, y \in X$.
- (i) Each of topological equivalence or Lipschitz equivalence is an equivalence relation on the set of metrics on a space X . This is easy to check.

- (ii) If d and d' are Lipschitz equivalent, then d and d' are topologically equivalent. This is easy to check. However, the other direction is false.

Example 3.17

Take $X = \mathbb{R}$, $d(x, y) = |x - y|$ and $d'(x, y) = \min\{1, |x - y|\}$. These are topologically equivalent since the open balls with small radii are the same. These are not, however, Lipschitz equivalent. Taking $x_n = n$, we have $d(x_n, 0) = n \rightarrow \infty$ but $d'(x_n, 0) = 1$ for $n \geq 1$. So, there is no $a > 0$ such that $a d(x, 0) \leq d'(x, 0)$ for all $x \in X$.

- (iii) A notion/property concerning a metric space X is a topological notion/property if it depends only on the topology on X and not on the specific metric generating the topology.
- (a) The convergence of sequences in X is a topological property, because, as one can easily check, for any (X, d) , $x_k \rightarrow x$ as defined in d is true if and only if for every open set $V \subseteq X$ with $x \in V$, we have that $x_k \in V$ for sufficiently large k .
 - (b) The boundedness of X is not a topological property.
 - (c) The compactness of X is a topological property.
 - (d) The completeness of X is not a topological property.

Definition 3.17 – Lipschitz equivalent norms

Two norms $\|\cdot\|$ and $\|\cdot\|'$ on a vector space V are Lipschitz equivalent if there exist $a, b > 0$ such that

$$a\|x\| \leq \|x\|' \leq b\|x\|$$

for all $x \in V$.

We adopt an enhanced notation for open balls, write $B_R^{\|\cdot\|}(0) = \{x \in V : \|x\| < R\}$.

Proposition 3.8

$\|\cdot\|$ and $\|\cdot\|'$ are Lipschitz equivalent if and only if there exist $r, R > 0$ such that

$$B_r^{\|\cdot\|}(0) \subseteq B_1^{\|\cdot\|'}(0) \subseteq B_R^{\|\cdot\|}(0).$$

Proof. On example sheet 3. □

On $\mathbb{R}^n$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ are Lipschitz equivalent.

Theorem 3.15

Any two norms on a finite-dimensional real vector space V are Lipschitz equivalent.

Proof. Let $\dim(V) = n$ and fix $\{e_1, \dots, e_n\}$ to be a basis for V . Define the Euclidean norm $\|\cdot\|_2$ on V by setting, for $x = \sum_{j=1}^n x_j e_j$, $\|x\|_2 = (\sum_{j=1}^n x_j^2)^{1/2}$. This is well-defined since x_j are uniquely determined by x . It is also easy to check that $\|\cdot\|_2$ is a norm. Since Lipschitz equivalence is an equivalence relation, it suffices to

prove that any other given norm $\|\cdot\|$ is Lipschitz equivalent to $\|\cdot\|_2$. Fix $\|\cdot\|$, note that for $x = \sum_{j=1}^n x_j e_j$,

$$\begin{aligned}\|x\| &= \left\| \sum_{j=1}^n x_j e_j \right\| \\ &\leq \sum_{j=1}^n \|x_j e_j\| \\ &= \sum_{j=1}^n |x_j| \|e_j\| \\ &\leq \left(\sum_{j=1}^n |x_j|^2 \right)^{1/2} \underbrace{\left(\sum_{j=1}^n \|e_j\|^2 \right)^{1/2}}_{\text{set this as } b} \text{ by Cauchy-Schwarz.}\end{aligned}$$

So, $\|x\| \leq b\|x\|_2$ for all x .

Lemma 3.1

The map $x \mapsto \|x\|$ is continuous on the space $(V, \|\cdot\|_2)$.

Proof. By the triangle inequality for $\|\cdot\|$, we have $|\|x\| - \|y\|| \leq \|x - y\| \leq b\|x - y\|_2$. The map $x \mapsto \|x\|$ is continuous and in fact Lipschitz. $\square$

Lemma 3.2

The unit sphere $S = \{x \in V : \|x\|_2 = 1\}$ of $(V, \|\cdot\|_2)$ is compact relative to $\|\cdot\|_2$.

Proof. Take a sequence $(x^{(k)})$ in S . Then, $x^{(k)} = \sum_{j=1}^n x_j^{(k)} e_j$ where $x_j^{(k)} \in \mathbb{R}$. Also, $\sum_{j=1}^n |x_j^{(k)}|^2 = 1$ for all k so $\tilde{x}^{(k)} = (x_1^{(k)}, \dots, x_n^{(k)})$ is bounded in $(\mathbb{R}^n, \|\cdot\|_2)$. By Bolzano-Weierstrass in $\mathbb{R}^n$, $(\tilde{x}^{(k)})$ has a subsequence $(\tilde{x}^{(k_j)})$ such that $\tilde{x}^{(k_j)} \rightarrow \tilde{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$. If we set $x = \sum_{j=1}^n x_j e_j \in V$, then $x^{(k_j)} \rightarrow x$ in $(V, \|\cdot\|_2)$. Hence, S is compact. $\square$

By these two lemmas and Theorem 3.14, there exist $x_0 \in S$ such that $\|x_0\| = \inf_{x \in S} \|x\|$. Also, $\|x_0\| > 0$ (otherwise $x_0 = 0$ but $0 \notin S$). Set $a = \|x_0\|$, so we have $\|x\| \geq a$ for all $x \in S$. For any $0 \neq x \in V$,

$$\left\| \frac{x}{\|x\|_2} \right\| \geq a \implies \|x\| \geq a\|x\|_2$$

for all $x \in V$. Hence, we have shown that $a\|x\|_2 \leq \|x\| \leq b\|x\|_2$ for all $x \in V$. $\square$

The Lipschitz equivalence of two norms $\|\cdot\|$ and $\|\cdot\|'$ on V is the statement that the two identity maps $i : (V, \|\cdot\|) \rightarrow (V, \|\cdot\|')$ and $i' : (V, \|\cdot\|') \rightarrow (V, \|\cdot\|)$ with $i(x) = x$ and $i'(x) = x$ are both Lipschitz.

Example 3.18

In infinite-dimensional vector spaces, inequivalent norms exist. $l_1 \subseteq l_\infty$ since $\sum_{j=1}^n |x_j| < \infty$ implies $x_j \rightarrow 0$ so $|x_j|$ is bounded. $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on l_1 are inequivalent norms. Take $x^{(k)} = (1, 1, \dots, 1, 0, 0, \dots)$ with k ones. $\|x^{(k)}\|_\infty = 1$ but $\|x^{(k)}\|_1 = k$.

In fact, the equivalence of any two norms in a real vector space characterises finite-dimensionality.

4 Differentiation in $\mathbb{R}^n$

In this section, we will always take the norm $\|\cdot\|$ on $\mathbb{R}^n$ to be the Euclidean norm. First, we need the following slightly different definition of a limit.

Definition 4.1 – Limit

Let $E \subseteq \mathbb{R}^n$ and $a \in \mathbb{R}^n$ be a limit point of E . If $f : E \rightarrow \mathbb{R}^m$ and $b \in \mathbb{R}^m$, then $\lim_{x \rightarrow a} f(x) = b$ means

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } x \in E \text{ and } 0 < \|x - a\| < \delta \implies \|f(x) - b\| < \varepsilon.$$

Here, we only need $0 < \|x - a\| < \delta$ instead of $\|x - a\| < \delta$. We do not ask that f be defined at $x = a$.

Given some function $f : U \rightarrow \mathbb{R}^m$ where $U \subseteq \mathbb{R}^n$ and a point $a \in U$, how can we define the derivative of f at a ? If $n = m = 1$, then say $f : (b, c) \rightarrow \mathbb{R}$ and $a \in (b, c)$. We know that $f'(a) = \lim_{h \rightarrow 0} (f(a+h) - f(a))/h$. If $n \geq 2$, we cannot divide by $h \in \mathbb{R}^n$, so we might try to define the derivative as $\lim_{h \rightarrow 0} (f(a+h) - f(a))/\|h\|$. However, if we take $n = 1$ in this definition, we have $\lim_{h \rightarrow 0} (f(a+h) - f(a))/|h|$ but this is very different to what we originally have. Taking $f(x) = |x|$, we get $\lim_{h \rightarrow 0} (f(0+h) - f(0))/|h| = 1$ but $|x|$ should be non-differentiable at $x = 0$. For $f(x) = x$, $\lim_{h \rightarrow 0} (f(0+h) - f(0))/|h|$ does not even exist.

Looking at the $n = 1$ situation more closely,

$$\begin{aligned} f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} &\iff \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - hf'(a)}{h} = 0 \\ &\iff \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - hf'(a)}{|h|} = 0 \\ &\iff \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Ah}{|h|} = 0 \text{ for some } A \in \mathbb{R}. \end{aligned}$$

In the higher dimensional setting, we need $h \in \mathbb{R}^n$ and $Ah \in \mathbb{R}^m$.

Definition 4.2 – Derivative

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \rightarrow \mathbb{R}^m$ and $a \in U$. Then, we say that f is differentiable at a if there is a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Ah}{\|h\|} = 0.$$

We say that A is the derivative of f at a and denote it by $A = Df(a)$.

Since we assume U is open, there is a ball $B_r(a) \subseteq U$, so that $a+h \in U$ for any h with $\|h\|$ sufficiently small.

Proposition 4.1

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}^m$.

- (i) If f is differentiable at $a \in U$, then $Df(a)$ is unique.
- (ii) If f is differentiable at $a \in U$, then f is continuous at a .
- (iii) Write $f = (f_1, f_2, \dots, f_m)$ where $f_j : U \rightarrow \mathbb{R}$. f is differentiable at $a \in U$ if and only if f_j is differentiable at a for each j , and

$$Df(a) = \begin{bmatrix} Df_1(a) \\ \vdots \\ Df_m(a) \end{bmatrix}.$$

Proof.

(i) If $A, A' \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ are both derivatives of f at a , then

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Ah}{\|h\|} = 0 \text{ and } \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - A'h}{\|h\|} = 0.$$

Taking their difference,

$$\lim_{h \rightarrow 0} \frac{Ah - A'h}{\|h\|} = 0.$$

Fix $u \in \mathbb{R}^n$ and take $h = tu$ with $t \rightarrow 0$. So,

$$\lim_{h \rightarrow 0} \frac{A(tu) - A'(tu)}{t\|u\|} = \lim_{h \rightarrow 0} \frac{Au - A'u}{\|u\|} = 0$$

where we removed the absolute value from $|t|$ since the limit is zero. Hence, $Au - A'u = 0$. Since u is arbitrary, $A = A'$.

(ii)

$$\lim_{h \rightarrow 0} f(a+h) - f(a) - (Df(a))h = 0 \implies \lim_{h \rightarrow 0} f(a+h) = f(a)$$

so f is continuous.

(iii) For any $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ with $A = [A_1 \dots A_m]^\top$, we have

$$\frac{f(a+h) - f(a) - Ah}{\|h\|} \rightarrow 0 \iff \frac{f_j(a+h) - f_j(a) - A_j h}{\|h\|} \rightarrow 0 \text{ as } h \rightarrow 0 \text{ for all } j.$$

So, if f is differentiable at a , then the left hand side holds with $A = Df(a)$, so the right holds with $A_j = Df_j(a)$. Hence, f_j is differentiable at a with $Df(a) = [Df_1(a) \dots Df_m(a)]^\top$. If all the f_j are differentiable at a , then the right hand side holds with $A_j = Df_j(a)$, so the left holds. That is, f is differentiable with $Df(a) = A$ and hence $Df(a) = [Df_1(a) \dots Df_m(a)]^\top$.

□

(iii) is useful as it allows us to reduce results to the case $m = 1$.

The differentiability of f at a point a means that the affine function $x \mapsto f(a) + (Df(a))(x - a)$ is a good approximation of f near a since

$$\lim_{x \rightarrow a} \frac{f(x) - [f(a) + (Df(a))(x - a)]}{\|x - a\|} = 0.$$

The deviation of $f(x)$ from $f(a) + (Df(a))(x - a)$ tends to zero faster than x tends to a .

We will write $o(x)$ for $x \in \mathbb{R}^n$ to denote any function with the property that $o(x)/\|x\| \rightarrow 0$ as $x \rightarrow 0$. So in this notation, differentiability of f at a is the statement $f(a+h) = f(a) + (Df(a))h + o(h)$.

Definition 4.3 – Directional derivative

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}^m$. Let $a \in U$, the directional derivative of f at a in the direction u (where $u \in \mathbb{R}^n$ is fixed) is

$$D_u f(a) = \lim_{t \rightarrow 0} \frac{f(a+tu) - f(a)}{t}$$

if the limit exists. This is just $[d/dt g(t)]_{t=0}$ where $g(t) = f(a+tu)$.

Proposition 4.2

If f is differentiable at $a \in U$, then for any $u \in \mathbb{R}^n$, the directional derivative $D_u f(a)$ exists, and is equal to $(Df(a))(u)$. In particular, the map $u \mapsto D_u f(a)$ is linear.

Proof.

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - (Df(a))(h)}{\|h\|} = 0 \\ \implies & \lim_{t \rightarrow 0} \frac{f(a+tu) - f(a) - t(Df(a))(u)}{t\|u\|} = 0 \text{ by taking } h = tu \text{ with } t \rightarrow 0 \\ \implies & \lim_{t \rightarrow 0} \frac{f(a+tu) - f(a)}{t} = (Df(a))(u) \end{aligned}$$

□

Partial derivatives are a special case of this. We usually take $m = 1$. Let e_j be the standard basis for $\mathbb{R}^n$. Take $u = e_j$, then

$$D_{e_j} f(a) = \lim_{t \rightarrow 0} \frac{f(a + te_j) - f(a)}{t}$$

is the j -th partial derivative of f at a if the limit exists. We use the notation $\partial f(a)/\partial x_j$ or $D_j f(a)$. So if $f : U \rightarrow \mathbb{R}$ is differentiable at $a \in U$, then the partial derivatives $D_j f(a)$ all exist, and $Df(a) = [D_1 f(a) \ \dots \ D_n f(a)]$. More generally,

Proposition 4.3

If $U \subseteq \mathbb{R}^n$ is open and $f : U \rightarrow \mathbb{R}^m$ is differentiable at $a \in U$, then the partial derivatives $D_j f_i(a)$ all exist, and the matrix for $Df(a)$ is given by $A = (a_{ij}) = (D_j f_i(a))$.

Proof. Combine the previous discussion with Proposition 4.1 (iii). □

In the case $m = 1$, $Df(a) = (\partial f(a)/\partial x_1, \dots, \partial f(a)/\partial x_n)$ is often called the gradient of f at a . We saw that differentiability at a implies the existence of all directional derivatives at a , which implies the existence of all partial derivatives at a . The converse is false for both.

Example 4.1

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(x, y) = \begin{cases} 0 & \text{if } xy = 0 \\ 1 & \text{otherwise.} \end{cases}$$

We have $(\partial f/\partial x)(0, 0) = (\partial f/\partial y)(0, 0) = 0$. For $u = (1, 1)$,

$$\frac{f(0+tu) - f(0)}{t} = \frac{f(t, t) - f(0)}{t} = \frac{1}{t}$$

so $\lim_{t \rightarrow 0} (f(0+tu) - f(0))/t$ does not exist. Hence, the existence of all partial derivatives does not imply the existence of directional derivatives.

Example 4.2

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(x, y) = \begin{cases} x^3/y & \text{if } y \neq 0 \\ 0 & \text{if } y = 0. \end{cases}$$

Then for $u = (u_1, u_2)$ and $t \neq 0$,

$$\frac{f(0 + tu) - f(0)}{t} = \begin{cases} tu_1^3/u_2 & \text{if } u_2 \neq 0 \\ 0 & \text{if } u_2 = 0 \end{cases}$$

so

$$D_u f(0) = \lim_{t \rightarrow 0} \frac{f(0 + tu) - f(0)}{t} = 0$$

and all directional derivatives exist and are zero. f is not differentiable at 0 since it is not continuous at 0 as $f(\delta, \delta^5) = 1/\delta^2 \rightarrow \infty$ as $\delta \rightarrow 0$. Hence, the existence of all directional derivatives does not imply differentiability.

Example 4.3

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(x, y) = \begin{cases} x^3/(x^2 + y^2) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Then, $|f(x, y)| \leq |x|$ so f is continuous at $(0, 0)$. By direct calculation, we have $(\partial f / \partial x)(0, 0) = 1$ and $(\partial f / \partial y)(0, 0) = 0$. In fact, for any $u = (u_1, u_2)$,

$$\frac{f(0 + tu) - f(0)}{t} = \frac{u_1^3}{u_1^2 + u_2^2}$$

so $D_u f(0)$ exists for all $u \neq 0$. But, $D_u f(0)$ does not depend linearly on u so f is not differentiable at 0. If it were, then $D_u f(0) = (Df(0))(u)$ which is linear in u . Alternatively, if f were differentiable at 0, then

$$(Df(0, 0))(h) = \begin{pmatrix} \frac{\partial f}{\partial x}(0) & \frac{\partial f}{\partial y}(0) \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = h_1$$

so

$$\frac{f(0 + h) - f(0) - (Df(0))(h)}{\|h\|} = \frac{h_1^3/(h_1^2 + h_2^2) - 0 - h_1}{(h_1^2 + h_2^2)^{1/2}} = -\frac{h_1 h_2^2}{(h_1^2 + h_2^2)^{3/2}} \not\rightarrow 0$$

as $h \rightarrow 0$. Hence, f is not differentiable at 0.

Theorem 4.1 – Continuity of partial derivatives implies differentiability

Let $f : U \rightarrow \mathbb{R}^m$ with $U \subseteq \mathbb{R}^n$ open and $a \in U$. Suppose that for some $B_r(a) \subseteq U$, the partial derivatives $D_j f_i(x)$ exist for every $x \in B_r(a)$. Suppose also that $D_j f_i$ is continuous at a for every i, j . Then, f is differentiable at a .

Proof. By Proposition 4.1 (iii), we can assume that $m = 1$ without loss of generality. Let $h = (h_1, \dots, h_n) \in \mathbb{R}^n$ and e_j be the standard unit vectors. Then,

$$f(a + h) - f(a) = \sum_{j=1}^n [f(a + h_1 e_1 + \dots + h_j e_j) - f(a + h_1 e_1 + \dots + h_{j-1} e_{j-1})].$$

By applying the mean value theorem to the function $g(t) = f(a + h_1 e_1 + \dots + h_{j-1} e_{j-1} + t e_j)$ with $t \in [0, h_j]$, we get that

$$f(a + h) - f(a) = \sum_{j=1}^n h_j D_j f(a + h_1 e_1 + \dots + h_{j-1} e_{j-1} + \theta_j h_j e_j)$$

for some $\theta_j \in (0, 1)$. So,

$$f(a+h) - f(a) - \sum_{j=1}^n h_j D_j f(a) = \sum_{j=1}^n h_j \underbrace{(D_j f(a + h_1 e_1 + \cdots + h_{j-1} e_{j-1} + \theta_j h_j e_j) - D_j f(a))}_{\rightarrow 0 \text{ as } h \rightarrow 0 \text{ since } D_j f \text{ is continuous at } a \text{ by hypothesis}}.$$

Hence, f is differentiable at a . □

For some of the next results, we need a convenient norm on the vector space $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) = \{A : \mathbb{R}^n \rightarrow \mathbb{R}^m \mid A \text{ linear}\}$. We will use the operator norm on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. Any $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous and, in fact, Lipschitz. Let $S = \{x \in \mathbb{R}^n : \|x\| = 1\}$ which is compact since it is a closed and bounded subset of $\mathbb{R}^n$. So, A is bounded on S . Define

$$\|A\|_{\text{op}} = \sup_{\|x\|=1} \|A(x)\| = \sup_{0 \neq x \in \mathbb{R}^n} \frac{\|A(x)\|}{\|x\|}.$$

(i) $\|A\|_{\text{op}}$ is a norm on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$.

(ii) For all $x \in \mathbb{R}^n$, $\|A(x)\| \leq \|A\|_{\text{op}} \|x\|$.

(iii) If $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and $B \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$, then $BA \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^p)$ and $\|BA\|_{\text{op}} \leq \|B\|_{\text{op}} \|A\|_{\text{op}}$.

Since $\dim(\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)) = mn < \infty$, any norm on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is equivalent to $\|\cdot\|_{\text{op}}$. For example, the obvious matrix norm $\|A\| = (\sum_{i,j} a_{ij}^2)^{1/2}$. But, because of (ii) and (iii), $\|\cdot\|_{\text{op}}$ is very convenient to use.

Proposition 4.4

(i) If $A \in \mathcal{L}(\mathbb{R}, \mathbb{R}^m)$, then $A(x) = xa$ for some fixed $a \in \mathbb{R}^m$ and $\|A\|_{\text{op}} = \|a\|_2$.

(ii) If $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$, then $A(x) = x \cdot a$ for some fixed $a \in \mathbb{R}^n$ and $\|A\|_{\text{op}} = \|a\|_2$.

(iii) If (A_{ij}) is the matrix of $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ with respect to the standard basis for $\mathbb{R}^n$ and $\mathbb{R}^m$, then we have

$$\frac{1}{\sqrt{n}} \left(\sum_{i,j} A_{ij}^2 \right)^{1/2} \leq \|A\|_{\text{op}} \leq \left(\sum_{i,j} A_{ij}^2 \right)^{1/2}.$$

Proof. On example sheet 4. □

Theorem 4.2 – Chain rule

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}^m$. Suppose that f is differentiable at $a \in U$. Let $V \subseteq \mathbb{R}^m$ be open with $f(U) \subseteq V$ and $g : V \rightarrow \mathbb{R}^p$. Suppose that g is differentiable at $f(a)$. Then, $g \circ f : U \rightarrow \mathbb{R}^p$ is differentiable at a and $D(g \circ f)(a) = (Dg(f(a))) \circ (Df(a))$.

Proof. Let $A = Df(a)$ and $B = Dg(f(a))$. We know

$$f(a+h) = f(a) + Ah + o(h) \text{ and } g(f(a) + k) = g(f(a)) + Bk + o(k).$$

Then,

$$\begin{aligned} (g \circ f)(a+h) &= g(f(a+h)) \\ &= g(f(a) + Ah + o(h)) \\ &= g(f(a)) + B(Ah + o(h)) + o(Ah + o(h)) \\ &= (g \circ f)(a) + BAh + B(o(h)) + o(Ah + o(h)). \end{aligned}$$

$\|B(o(h))\| \leq \|B\|_{\text{op}}\|o(h)\|$ so $B(o(h)) = o(h)$. Also, for small $\|h\|$,

$$\|Ah + o(h)\| \leq \|A\|_{\text{op}}\|h\| + \|o(h)\| \leq (\|A\|_{\text{op}} + 1)\|h\|.$$

Also, $o(Ah + o(h)) = \varepsilon(\|h\|)\|Ah + o(h)\|$ with $\varepsilon(t) \rightarrow 0$ as $t \rightarrow 0^+$. So, $o(Ah + o(h)) = o(h)$. Hence, we have $(g \circ f)(a + h) = (g \circ f)(a) + BAh + o(h)$ so $g \circ f$ is differentiable at a with $D(g \circ f)(a) = BA$. $\square$

In 1D calculus, we have the mean value theorem for maps $f : [a, b] \rightarrow \mathbb{R}$. That is, if $f : [a, b] \rightarrow \mathbb{R}$ is continuous and if f is differentiable in (a, b) , then $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$. To what extent does this extend to maps $f : U \rightarrow \mathbb{R}^m$ with $U \subseteq \mathbb{R}^n$? Even when $n = 1$, the mean value theorem does not directly extend to the case $m \geq 2$. There need not be $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$.

Theorem 4.3 – Mean value inequality with $n = 1$

Let $f : [a, b] \rightarrow \mathbb{R}^m$ be continuous and assume that f is differentiable on (a, b) . Suppose that $\|Df(t)\|_{\text{op}} \leq M$ for some constant M and all $t \in (a, b)$. Then,

$$\|f(b) - f(a)\| \leq M(b - a).$$

Proof. Let $v = f(b) - f(a) = (v_1, \dots, v_m)$. Let $g(t) = v \cdot f(t) = \sum_{i=1}^m v_i f_i(t)$. $g : [a, b] \rightarrow \mathbb{R}$ is continuous and differentiable on (a, b) . By the mean value theorem, $g(b) - g(a) = g'(c)(b - a)$ for some $c \in (a, b)$. $g(b) - g(a) = v \cdot (f(b) - f(a)) = \|v\|^2$ and

$$g'(t) = \sum_{i=1}^m v_i f'_i(t) \leq \|v\| \|f'(t)\| = \|v\| \|Df(t)\|_{\text{op}} \leq M \|v\|.$$

If $\|v\| = 0$, then $f(b) = f(a)$ and there is nothing further to prove. Else,

$$\|v\|^2 = g(b) - g(a) \leq |g'(c)|(b - a) \leq M \|v\| (b - a)$$

so $\|v\| \leq M(b - a)$. $\square$

Theorem 4.4

Let $B_r(a) \subseteq \mathbb{R}^n$ and $f : B_r(a) \rightarrow \mathbb{R}^m$. Suppose that f is differentiable at x for each $x \in B_r(a)$. Suppose further that $\|Df(x)\|_{\text{op}} \leq M$ for some constant M and all $x \in B_r(a)$. Then, for any $b_1, b_2 \in B_r(a)$, we have

$$\|f(b_2) - f(b_1)\| \leq M \|b_2 - b_1\|.$$

Proof. Let $g(t) = f(tb_1 + (1 - t)b_2)$ for $t \in [0, 1]$. Then, $g : [0, 1] \rightarrow \mathbb{R}^m$ is continuous and differentiable on $(0, 1)$. $Dg(t) = (Df(tb_1 + (1 - t)b_2))(b_1 - b_2)$ so

$$\|Dg(t)\|_{\text{op}} \leq \|Df(tb_1 + (1 - t)b_2)\|_{\text{op}} \|b_1 - b_2\| \leq M \|b_1 - b_2\|.$$

Applying Theorem 4.3 with $f = g$, $a = 0$, $b = 1$ and $M \|b_1 - b_2\|$ in place of M , we get the result. $\square$

The above proof carries over to maps $f : U \rightarrow \mathbb{R}^m$ where $U \subseteq \mathbb{R}^n$ is a convex set.

Corollary 4.1

If $f : B_r(a) \rightarrow \mathbb{R}^m$ is differentiable with $Df(x) = 0$ for all $x \in B_r(a)$, then f is constant.

Proof. Apply Theorem 4.4 with $M = 0$. $\square$

For general open sets $U \subseteq \mathbb{R}^n$ and maps $f : U \rightarrow \mathbb{R}^m$, $Df = 0$ does not imply that f is constant. For

example, let $U = (0, 1) \cup (2, 3)$ and

$$f(x) = \begin{cases} 1 & \text{if } x \in (0, 1) \\ 2 & \text{if } x \in (2, 3) \end{cases}$$

but $f'(x) = 0$ everywhere on U . We need U to be, additionally, path-connected to extend Corollary 4.1.

Definition 4.4 – Path connectedness

A subset $E \subseteq \mathbb{R}^n$ is path connected if for any two points $a, b \in E$, there is a continuous map $\gamma : [0, 1] \rightarrow E$ with $\gamma(0) = a$ and $\gamma(1) = b$.

Theorem 4.5

Let $U \subseteq \mathbb{R}^n$ be open and path connected. If $f : U \rightarrow \mathbb{R}^m$ is differentiable with $Df(x) = 0$ for all $x \in U$, then f is constant.

Proof. By considering the component functions $f_1, \dots, f_n$, we can simply consider the case when $m = 1$. Let $a, b \in U$ be any two points. There exists a continuous $\gamma : [0, 1] \rightarrow U$ such that $\gamma(0) = a$ and $\gamma(1) = b$. Let $g(t) = f(\gamma(t))$ with $t \in [0, 1]$. We need to show $f(a) = f(b)$ which is equivalent to $g(0) = g(1)$. Since γ need not be differentiable, we cannot simply differentiate g .

For $s \in (0, 1)$, $\gamma(s) \in U$. Since U is open, there exists $\varepsilon > 0$ such that $B_\varepsilon(\gamma(s)) \subseteq U$. Since γ is continuous, there exists $\delta > 0$ such that $\gamma(s - \delta, s + \delta) \subseteq B_\varepsilon(\gamma(s))$. Since $Df = 0$, by Corollary 4.1, f is constant in $B_\varepsilon(\gamma(s))$ so g is constant in $(s - \delta, s + \delta)$. Hence, g is differentiable in $(s - \delta, s + \delta)$ with $g'(t) = 0$ for all $t \in (s - \delta, s + \delta)$. In particular, $g'(s) = 0$. Since s is arbitrary, $g'(t) = 0$ for all $t \in (0, 1)$. Apply the mean value theorem to get $f(b) - f(a) = g(1) - g(0) = 0$. $\square$

Definition 4.5 – C^k functions

Let $U \subseteq \mathbb{R}^n$ be open. A function $f : U \rightarrow \mathbb{R}$ is said to be C^1 on U if all partial derivatives $D_j f$ exist in U and are continuous on U .

Similarly, f is C^2 on U if $D_j f : U \rightarrow \mathbb{R}$ is C^1 for each j . That is, the second partial derivatives $D_i D_j f$ all exist in U and are continuous on U .

It turns out that if f is C^2 , then $D_i D_j f(x) = D_j D_i f(x)$ for all $i, j \in \{1, \dots, n\}$ and all $x \in U$. In fact, we have the following stronger result.

Theorem 4.6

Let $f : U \rightarrow \mathbb{R}$ with $U \subseteq \mathbb{R}^n$ open. Let $a \in U$ so $B_r(a) \subseteq U$ for some r . Fix $i, j \in \{1, \dots, n\}$ and assume that the partial derivatives $D_i D_j f(x)$ and $D_j D_i f(x)$ exist for all $x \in B_r(a)$ and are continuous at $x = a$. Then,

$$D_i D_j f(x) = D_j D_i f(x).$$

Proof. If $i = j$, then there is nothing to prove. So, assume $i \neq j$. Let

$$g_{ij}(t) = f(a + te_i + te_j) - f(a + te_i) - f(a + te_j) + f(a)$$

for $t \in (0, r/2)$. Then,

$$\begin{aligned} g_{ij}(t) &= \varphi(1) - \varphi(0) \text{ where } \varphi(s) = f(a + ste_i + te_j) - f(a + ste_i) \\ &= \varphi'(\theta) \text{ for some } \theta \in (0, 1) \text{ by the mean value theorem} \\ &= t(D_i f(a + \theta te_i + te_j) - D_i f(a + \theta te_i)) \\ &= t^2 D_j D_i f(a + \theta te_i + \eta te_j) \text{ for some } \eta \in (0, 1) \text{ by applying the MVT to } s \mapsto D_i f(a + \theta te_i + ste_j). \end{aligned}$$

We can repeat the argument switching i and j to get $g_{ji}(t) = t^2 D_i D_j f(a + \tilde{\theta} te_i + \tilde{\eta} te_j)$ for some $\tilde{\theta}, \tilde{\eta} \in (0, 1)$. Since $g_{ij}(t) = g_{ji}(t)$, we get

$$D_j D_i f(a + \theta te_i + \eta te_j) = D_i D_j f(a + \tilde{\theta} te_i + \tilde{\eta} te_j).$$

Let $t \rightarrow 0$ and we get the result by the continuity of the second partial derivatives from assumption. $\square$

By repeatedly applying this, we get that if f is C^k on U for some $k \in \mathbb{N}$, that is, all partial derivatives of order k , $D_{i_1} \dots D_{i_k} f(x)$, exist and are continuous for any choice of $i_1, \dots, i_k \in \{1, \dots, n\}$. Then, the order of differentiation does not matter, that is,

$$D_{i_1} \dots D_{i_k} f(x) = D_{i_{j_1}} \dots D_{i_{j_k}} f(x)$$

for all $x \in U$ and for any permutation $j_1, \dots, j_k$ of $1, \dots, k$.

What is the derivative of Df ? If $f : U \rightarrow \mathbb{R}^m$, with $U \subseteq \mathbb{R}^n$ open, is differentiable on U , then Df is a map $Df : U \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$ with components $D_j f_k$. Df is differentiable at $a \in U$ if and only if $D_j f_k$ are all differentiable at a . If so, $D(Df)(a)$ is the unique linear transformation $D(Df)(a) : \mathbb{R}^n \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ satisfying

$$\lim_{h \rightarrow 0} \frac{Df(a+h) - Df(a) - (D(Df)(a))(h)}{\|h\|} = 0 \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m).$$

The limit is interpreted to be taken with respect to say the operator norm in $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$.

From linear algebra, suppose $\varphi : \mathbb{R}^l \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is a linear transformation. Define $\Phi = \mathbb{R}^l \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $\Phi(u, v) = (\varphi(u))(v)$. Then, Φ is bilinear, that is $\Phi(au + bv, w) = a\Phi(u, w) + b\Phi(v, w)$ and $\Phi(u, av + bw) = a\Phi(u, v) + b\Phi(u, w)$. Conversely, if $\Phi : \mathbb{R}^l \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ is bilinear, then the map $\varphi : \mathbb{R}^l \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ defined by $\varphi(u) = (v \mapsto \Phi(u, v))$ is linear. If $m = 1$, we say that Φ is a bilinear form.

Definition 4.6

Suppose Df is differentiable at $x = a$. Define $D^2 f(a) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ by

$$(D^2 f(a))(u, v) = ((D(Df)(a))(u))(v).$$

Theorem 4.7

Suppose $f : U \rightarrow \mathbb{R}^m$ is C^2 on U , that is, f_k is C^2 on U for all k . Then, $D^2 f(a)$ is a symmetric bilinear map. Moreover, for $u = \sum_{i=1}^n u_i e_i$ and $v = \sum_{i=1}^n v_i e_i$ with e_i being the standard basis, we have

$$(D^2 f(a))(u, v) = \left(\sum_{i=1}^n \sum_{j=1}^n D_{ij} f_k(a) u_i v_j \right)_{1 \leq k \leq m} \in \mathbb{R}^m.$$

Proof. Since f is C^2 , the components $D_j f_k$ of Df have continuous partial derivatives, so $D_j f_k$ are each differ-

entiable. Hence, Df is differentiable and $D^2f(a)$ exists. Using bilinearity,

$$(D^2f(a))(u, v) = \sum_{i=1}^n \sum_{j=1}^n (D^2f(a))(e_i, e_j) u_i v_j.$$

By the definition of $D(Df)(a)$,

$$\lim_{h \rightarrow 0} \frac{Df(a+h) - Df(a) - (D(Df)(a))(h)}{\|h\|} = 0 \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m).$$

Taking $h = te_i$,

$$\lim_{t \rightarrow 0} \frac{Df(a+te_i) - Df(a) - t(D(Df)(a))(e_i)}{t} = 0.$$

Have this act on e_j and get

$$\lim_{t \rightarrow 0} \frac{(Df(a+te_i))(e_j) - (Df(a))(e_j) - t((D(Df)(a))(e_i))(e_j)}{t} = 0$$

so

$$((D(Df)(a))(e_i))(e_j) = \lim_{t \rightarrow 0} \frac{D_j f(a+te_i) - D_j f(a)}{t} = D_i D_j f(a).$$

Hence, $(D^2f(a))(e_i, e_j) = D_{ij}f(a)$ which gives the result. $\square$

In the case of $m = 1$, we can use the bilinear form $D^2f(a)$ to analyse the critical points of f , which are the points where $Df(a) = 0$.

Some facts from linear algebra. Let $B : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a symmetric bilinear form, so

$$B(u, v) = B(v, u) = \sum_{i,j=1}^n a_{ij} u_i v_j$$

for a symmetric matrix (a_{ij}) with $a_{ij} \in \mathbb{R}$. Associated to this is the function

$$Q(u) = B(u, u) = \sum_{i,j=1}^n a_{ij} u_i u_j,$$

a quadratic form. It is a homogeneous degree two polynomial in the variables $\{u_1, \dots, u_n\}$. We can recover B from Q by the formula $B(u+v, u+v) = Q(u) + Q(v) + 2B(u, v)$. Hence, there is a bijection between symmetric bilinear forms and quadratic forms.

- (i) Q is said to be positive definite if $Q(u) > 0$ for all $u \in \mathbb{R}^n \setminus \{0\}$.
- (ii) Q is said to be positive semi-definite if $Q(u) \geq 0$ for all $u \in \mathbb{R}^n$.
- (iii) Q is said to be negative definite if $Q(u) < 0$ for all $u \in \mathbb{R}^n \setminus \{0\}$.
- (iv) Q is said to be negative semi-definite if $Q(u) \leq 0$ for all $u \in \mathbb{R}^n$.

In the case that $B = D^2f(a)$ for a C^2 function $f : U \rightarrow \mathbb{R}$ and $a \in U$. The associated quadratic form Q is the Hessian form, denoted

$$H_{f,a}(u) = \sum_{i,j=1}^n D_{ij}f(a) u_i u_j = u^\top A u$$

where $A = (D_{ij}f(a))$.

Definition 4.7 – Critical points and (strict) local minima/maxima

Let $f : U \rightarrow \mathbb{R}$ be C^2 and $a \in U$.

- (i) The point a is a critical point of f if $Df(a) = 0$.
- (ii) f has a local minimum at a if there exists $r > 0$ such that $f(x) \geq f(a)$ for all $x \in B_r(a)$.
- (iii) f has a strict local minimum at a if there exists $r > 0$ such that $f(x) > f(a)$ for all $x \in B_r(a) \setminus \{a\}$.

We can similarly define local maxima and strict local maxima.

Theorem 4.8

Let $f : U \rightarrow \mathbb{R}$, with $U \subseteq \mathbb{R}^n$ open, be a C^2 function and $a \in U$. Then,

- (i) we have the second order Taylor expansion

$$f(a+h) = f(a) + (Df(a))(h) + \frac{1}{2}(D^2f(a))(h,h) + E(h)$$

with $E(h) = o(h^2)$, that is $E(h)/\|h\|^2 \rightarrow 0$ as $h \rightarrow 0$.

- (ii) if a is a critical point of f and $H_{f,a}(h) = (D^2f(a))(h,h)$ is positive definite (or respectively, negative definite), then f has a strict local minimum (respectively, strict local maximum) at a .
- (iii) if f has a local minimum (or respectively, local maximum) at a , then $Df(a) = 0$ and $H_{f,a}(h)$ is positive semi-definite (respectively, negative semi-definite).

Proof.

- (i) If $g : [0, 1] \rightarrow \mathbb{R}$ is C^2 , then

$$\int_0^1 (1-t)g''(t) dt = g(1) - g(0) - g'(0)$$

by integrating by parts. Apply this with $g(t) = f(a+th)$, by the chain rule, $g'(t) = (Df(a+th))(h) = \sum_{j=1}^n h_j D_j f(a+th)$ and

$$g''(t) = \sum_{i,j=1}^n h_i h_j D_{ij} f(a+th) = \sum_{i,j=1}^n h_i h_j (D_{ij} f(a+th) - D_{ij} f(a)) + \underbrace{\sum_{i,j=1}^n h_i h_j D_{ij} f(a)}_{= (D^2 f(a))(h,h)}.$$

So,

$$\begin{aligned} f(a+h) &= g(1) \\ &= g(0) + g'(0) + \int_0^1 (1-t)g''(t) dt \\ &= f(a) + (Df(a))(h) + \underbrace{\sum_{i,j=1}^n \left[h_i h_j \int_0^1 (1-t)(D_{ij} f(a+th) - D_{ij} f(a)) dt \right]}_{\text{call this } E(h)} + (D^2 f(a))(h,h) \int_0^1 (1-t) dt \\ &= f(a) + (Df(a))(h) + \frac{1}{2}(D^2 f(a))(h,h) + E(h). \end{aligned}$$

Also,

$$\frac{|E(h)|}{\|h\|^2} \leq \sum_{i,j=1}^n \int_0^1 (1-t) |D_{ij} f(a+th) - D_{ij} f(a)| dt.$$

Given $\varepsilon > 0$, there exists $\delta > 0$ such that $h \in B_\delta(0)$ implies $|D_{ij} f(a+th) - D_{ij} f(a)| < \varepsilon$ for all $t \in [0, 1]$ and for all i, j . Hence, $h \in B_\delta(0)$ implies $|E(h)|/\|h\|^2 < \varepsilon/2$ so $|E(h)|/\|h\|^2 \rightarrow 0$ as $h \rightarrow 0$.

- (ii) Suppose $Df(a) = 0$ and $H_{f,a}(h) > 0$ for all $h \in \mathbb{R}^n \setminus \{0\}$. Let $\varepsilon_0 = \inf_{h \in S} H_{f,a}(h)$ where $S = \{x \in \mathbb{R}^n : \|x\| = 1\}$. Since $h \mapsto H_{f,a}(h)$ is continuous and S is compact, the infimum is attained so $\varepsilon_0 > 0$. So, for $h \in \mathbb{R}^n$, we have that $H_{f,a}(h) \geq \varepsilon_0 \|h\|^2$. Hence,

$$\begin{aligned} f(a+h) &\geq f(a) + (\varepsilon_0/2)\|h\|^2 + o(h^2) \\ &\geq f(a) + (\varepsilon_0/4)\|h\|^2 \text{ for all } h \in B_\delta(0) \text{ for some } \delta > 0 \\ &> f(a) \text{ if } h \in B_\delta(0) \setminus \{0\}. \end{aligned}$$

So, f has a strict local minimum at a . The argument for strict local maximum is similar.

- (iii) On example sheet 4.

□

From linear algebra, $H_{f,a}(h) = \sum_{i,j=1}^n D_{ij}f(a)h_ih_j$ is positive definite (or respectively, negative definite) if and only if the real symmetric matrix $(D_{ij}f(a))$ has real positive (respectively, real negative) eigenvalues.

5 The contraction mapping theorem and applications

Definition 5.1 – Contraction mapping

Let (X, d) be a metric space. A contraction mapping is a function $f : X \rightarrow X$ such that there exists λ with $0 \leq \lambda < 1$ such that

$$d(f(x), f(y)) \leq \lambda d(x, y)$$

for all $x, y \in X$.

If f is a contraction mapping, then f is Lipschitz with Lipschitz constant < 1 . In particular, a contraction mapping is uniformly continuous.

Theorem 5.1 – Contraction mapping theorem

If (X, d) is a non-empty complete metric space, and if $f : X \rightarrow X$ is a contraction mapping, then f has a unique fixed point $x \in X$, that is, $f(x) = x$ for some unique $x \in X$.

Proof. We first prove uniqueness. If $x = f(x)$ and $y = f(y)$ are both fixed points, then

$$d(x, y) = d(f(x), f(y)) \leq \lambda d(x, y)$$

where $\lambda \in [0, 1)$ is the contraction constant. Since $\lambda < 1$, we must have $d(x, y) = 0$ and hence $x = y$.

To see existence, pick any $x_0 \in X$. Define a sequence (x_n) by $x_{n+1} = f(x_n)$, so $x_n = f^n(x_0)$. Then,

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq \lambda d(x_n, x_{n-1}) \leq \cdots \leq \lambda^n d(x_1, x_0).$$

For $m > n$, by the triangle inequality,

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \leq \sum_{j=n}^{m-1} \lambda^j d(x_1, x_0) \leq \frac{\lambda^n}{1-\lambda} d(x_1, x_0)$$

so (x_n) is Cauchy. Since X is complete, there exists $x \in X$ such that $x_n \rightarrow x$. By continuity of f , $f(x_n) \rightarrow f(x)$ so $x_{n+1} \rightarrow f(x)$. Also, $x_{n+1} \rightarrow x$. By the uniqueness of limits, $x = f(x)$. □

Completeness of X is necessary. Consider $f : (0, 1] \rightarrow (0, 1]$ with $f(x) = x/2$. This is a contraction mapping

without a fixed point in $(0, 1]$.

Note that the above process is constructive. In particular, it provides the estimate

$$d(x, f^n(x_0)) \leq \frac{\lambda^n}{1 - \lambda} d(f(x_0), x_0).$$

This can be used to determine how many steps n are needed to approximate the fixed point x by $f^n(x)$ to achieve a given degree of accuracy.

Theorem 5.2 – Inverse function theorem

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}^n$ be a C^1 map. Let $a \in U$ and suppose that $Df(a) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ is invertible. Then, there exists an open set $V \subseteq U$ with $a \in V$ and another open set $W \subseteq \mathbb{R}^n$ with $f(a) \in W$ such that $f|_V : V \rightarrow W$ is a diffeomorphism, that is, a C^1 bijection with a C^1 inverse $g : W \rightarrow V$.

Proof. By considering $(Df(a))^{-1}f$ in place of f , we can assume without loss of generality that $Df(a) = I$, the identity map $\mathbb{R}^n \rightarrow \mathbb{R}^n$. Since f is C^1 , there exists $r > 0$ such that $\|Df(x) - I\|_{\text{op}} < 1/2$ for all $x \in B_r(a)$. This follows from the fact that if $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$, then $\|A\|_{\text{op}} \leq (\sum_{i,j=1}^n A_{ij}^2)^{1/2}$. Applying this to $A = Df(x) - I$,

$$\|Df(x) - I\|_{\text{op}} \leq \left[\sum_{i,j=1}^n (D_i f_j(x) - D_i f_j(a))^2 \right]^{1/2}$$

which could be made arbitrarily small given that x is sufficiently close to a by the continuity of Df .

We will complete the proof with $W = B_{r/2}(f(a))$ and $V = f^{-1}(W) \cap B_r(a)$. Clearly, W is open with $f(a) \in W$. By the continuity of f , $f^{-1}(W)$ is also open. Since the intersection of open sets is open, V is open, and $a \in V$. The proof is done in four steps.

(i) $f|_V : V \rightarrow W$ is a bijection.

The claim is equivalent to the statement that

$$\begin{aligned} & \text{for all } y \in W, \text{ there exists a unique } x \in V \text{ such that } f(x) = y \\ \iff & \text{for all } y \in W, \text{ there exists a unique } x \in V \text{ such that } \underbrace{y + x - f(x)}_{\text{call this } T(x)} = x \\ \iff & \text{for all } y \in W, \text{ the map } T : U \rightarrow \mathbb{R}^n \text{ has a unique fixed point.} \end{aligned}$$

Consider the restriction of T to $\overline{B_r(a)}$, that is $T : \overline{B_r(a)} \rightarrow \mathbb{R}^n$. We first check that T maps into $\overline{B_r(a)}$.

$$\begin{aligned} \|T(x) - a\| &= \|y + x - f(x) - a\| \\ &= \|x - f(x) - a + f(a) + y - f(a)\| \\ &\leq \|x - f(x) - (a - f(a))\| + \|y - f(a)\| \\ &\leq \frac{1}{2}\|x - a\| + \|y - f(a)\| \text{ by the mean value inequality since } \|Df(x) - I\|_{\text{op}} \leq \frac{1}{2} \\ &< \frac{r}{2} + \frac{r}{2} = r \end{aligned}$$

so T maps $\overline{B_r(a)}$ into $B_r(a) \subseteq \overline{B_r(a)}$. Also, T is a contraction map since for all $x_1, x_2 \in \overline{B_r(a)}$,

$$\|T(x_1) - T(x_2)\| = \|x_1 - f(x_1) - (x_2 - f(x_2))\| \leq (1/2)\|x_1 - x_2\|$$

by the mean value inequality as before. Moreover, $\overline{B_r(a)}$ is complete since it is a closed subset of the complete space $\mathbb{R}^n$. So, by the contraction mapping theorem, there exists a unique $x \in \overline{B_r(a)}$ such that $T(x) = x$. Since $T(x) \in B_r(a)$, we in fact have $x \in B_r(a)$. Also, $T(x) = x \iff f(x) = y$, and since $y \in W$, we have $x \in f^{-1}(W)$. So, $x \in f^{-1}(W) \cap B_r(a) = V$.

(ii) Let $g = f|_V^{-1} : W \rightarrow V$. We claim that

$$\|g(y_1) - g(y_2)\| \leq 2\|y_1 - y_2\|$$

for all $y_1, y_2 \in W$. That is, g is Lipschitz with $\text{Lip}(g) \leq 2$

For $y_1, y_2 \in W$, let $x_i = g(y_i)$.

$$\begin{aligned} \|x_1 - x_2\| &\leq \|x_1 - f(x_1) - (x_2 - f(x_2))\| + \|f(x_1) - f(x_2)\| \\ &\leq \frac{1}{2}\|x_1 - x_2\| + \|f(x_1) - f(x_2)\| \text{ by (i).} \end{aligned}$$

So, $\|x_1 - x_2\| \leq 2\|f(x_1) - f(x_2)\| \implies \|g(y_1) - g(y_2)\| \leq 2\|y_1 - y_2\|$.

(iii) $Df(x)$ is invertible for every $x \in \overline{B_r(a)}$. g is differentiable at each $y \in W$ with $Dg(y) = D(f(g(y)))^{-1}$.

For any $x \in \overline{B_r(a)}$ and $v \in \mathbb{R}^n$, if $(Df(x))(v) = 0$, then

$$\|v\| = \|(Df(x))(v) - v\| \leq \|Df(x) - I\|_{\text{op}}\|v\| \leq \frac{1}{2}\|v\|.$$

So, $\|v\| = 0 \implies v = 0$. Hence, $Df(x)$ is invertible. To check that g is differentiable, set $b \in W$. There exists a unique c such that $f(c) = b$.

$$\begin{aligned} \|g(b+k) - g(b) - \underbrace{(Df(c))^{-1}(k)}_{\text{call this } h}\| &= \|g(f(c) + (Df(c))(h)) - g(f(c))\| \\ &= \|g(f(c+h) + o(h)) - g(f(c+h))\| \\ &\leq 2\|o(h)\| \text{ by (ii)} \\ &= o(h) \end{aligned}$$

so g is differentiable with $Dg(b) = D(f(g(b)))^{-1}$.

(iv) Dg is continuous on W .

This is clear from (iii), since g is continuous, Df is continuous and the map $A \mapsto A^{-1}$ is continuous on $\text{GL}_n(\mathbb{R})$.

And we are done. □

We can think of $\mathbb{R}^n$ as $\mathbb{R}^{n-m} \times \mathbb{R}^m$ and write a point in $\mathbb{R}^n$ as (x, y) with $x \in \mathbb{R}^{n-m}$ and $y \in \mathbb{R}^m$. We can also write $Df(x, y) = (D_x f(x, y), D_y f(x, y))$.

Theorem 5.3 – Implicit function theorem

Let $U \subseteq \mathbb{R}^n$ be open. Let $m < n$ and $f : U \rightarrow \mathbb{R}^m$ be C^1 . Let $\Sigma = f^{-1}(0) = \{(x, y) \in U : f(x, y) = 0\}$. Suppose that $(a, b) \in \Sigma$ is such that $D_y f(a, b)$ is invertible. Then, there exists an open set $V \subseteq U$ with $(a, b) \in V$, and another set W of $\mathbb{R}^{n-m}$ with $a \in W$, and a C^1 map $g : W \rightarrow \mathbb{R}^m$ such that $\Sigma \cap V = \{(x, g(x)) : x \in W\}$.

Proof. Let $F(x, y) = (x, f(x, y))$ so $F : U \rightarrow \mathbb{R}^n$ is C^1 . Then,

$$DF = \begin{bmatrix} I_{n-m} & 0 \\ D_x f & D_y f \end{bmatrix}$$

and $DF(a, b)$ is invertible since $D_y f(a, b)$ is invertible. By the inverse function theorem, there exists an open $V \subseteq U$ with $(a, b) \in V$ such that $F : V \rightarrow F(V)$ is a bijection with C^1 inverse $G : F(V) \rightarrow V$. Write $G = (G_1, G_2)$. $F \circ G = \text{id}_{F(V)}$ implies that $(G_1(x, y), f(G_1(x, y), G_2(x, y))) = (x, y)$ for all $(x, y) \in F(V)$. So, $G_1(x, y) = x$ and

$$f(x, G_2(x, y)) = y \text{ for all } (x, y) \in F(V). \quad (1)$$

Also, $G \circ F = \text{id}_V$ implies that $(G_1(x, f(x, y)), G_2(x, f(x, y))) = (x, y)$ for all $(x, y) \in F(V)$. So,

$$G_2(x, f(x, y)) = y \text{ for all } (x, y) \in F(V). \quad (2)$$

Note that $F(a, b) = (a, f(a, b)) = (a, 0)$ so $(a, 0) \in F(V)$. Set $W = p(F(V) \cap (\mathbb{R}^{n-m} \times \{0\}))$ where $p : \mathbb{R}^n \rightarrow \mathbb{R}^{n-m}$ is the orthogonal projection. Since $F(V)$ is open in $\mathbb{R}^n$, it is clear that W is open in $\mathbb{R}^{n-m}$ with $a \in W$. Set $g : W \rightarrow \mathbb{R}^m$ defined by $g(x) = G_2(x, 0)$. From (1), $f(x, g(x)) = 0$ so $(x, g(x)) \in \Sigma$ and $\{(x, g(x)) : x \in W\} \subseteq \Sigma \cap V$. If $(x, y) \in \Sigma$, then $f(x, y) = 0$, and from (2), $g(x) = y$ so $\Sigma \cap V \subseteq \{(x, g(x)) : x \in W\}$. $\square$

Under the appropriate assumptions, the inverse function theorem says that given a point $b \in U = \text{domain}(f) \subseteq \mathbb{R}^m$, for any given $x = (x_1, \dots, x_m)$ near $f(b)$, the system of equations

$$\begin{aligned} f_1(y_1, \dots, y_m) &= x_1 \\ f_2(y_1, \dots, y_m) &= x_2 \\ &\vdots \\ f_m(y_1, \dots, y_m) &= x_m \end{aligned}$$

is uniquely solvable for $y = (y_1, \dots, y_m) \in \mathbb{R}^m$ near b . The implicit function theorem is the more general statement that given $(a, b) \in U = \text{domain}(f) \subseteq \mathbb{R}^{n-m} \times \mathbb{R}^m$ with $f(a, b) = 0$, for any given $x = (x_1, \dots, x_{n-m})$ near $a \in \mathbb{R}^{n-m}$, the system

$$\begin{aligned} f_1(x_1, \dots, x_{n-m}, y_1, \dots, y_m) &= 0 \\ &\vdots \\ f_m(x_1, \dots, x_{n-m}, y_1, \dots, y_m) &= 0 \end{aligned}$$

is uniquely solvable for $y = (y_1, \dots, y_m) \in \mathbb{R}^m$ near b . In particular, we can deduce the inverse function theorem by applying the implicit function theorem to the map $\tilde{f}(x, y) = f(y) - x$.

Our second application of the contraction mapping theorem proves the existence of solutions to systems of ordinary differential equations. We want to solve

$$\frac{df}{dt} = F(t, f(t))$$

subject to $f(t_0) = x_0$ where $F : [a, b] \times \overline{B_R(x_0)} \rightarrow \mathbb{R}^n$. This is a coupled system of ODEs.

$$\begin{aligned} df_1/dt &= F_1(t, f_1(t), \dots, f_n(t)) \\ &\vdots \\ df_n/dt &= F_n(t, f_1(t), \dots, f_n(t)) \end{aligned}$$

which can be very complicated. Nonetheless, we have the following general local existence theorem.

Theorem 5.4 – Picard-Lindelöf existence theorem

Let $x_0 \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$ with $a < b$. Let $R > 0$ and $t_0 \in [a, b]$. Let $F : [a, b] \times \overline{B_R(x_0)} \rightarrow \mathbb{R}^n$ be continuous. Suppose further that there exists $K \geq 0$ such that for all $t \in [a, b]$ and $x, y \in \overline{B_R(x_0)}$, we have

$$\|F(t, x) - F(t, y)\| \leq K\|x - y\|.$$

Then,

- (i) there exists $\varepsilon > 0$ and a unique differentiable function $f : [t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b] \rightarrow \mathbb{R}^n$ such that $df/dt = F(t, f(t))$ for all $t \in [t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b]$, and $f(t_0) = x_0$.
- (ii) if additionally,

$$\sup_{[a, b] \times \overline{B_R(x_0)}} \|F\| \leq R/(b - a),$$

then there exists a unique differentiable $f : [a, b] \rightarrow \mathbb{R}^n$ such that $df/dt = F(t, f(t))$ for all $t \in [a, b]$, and $f(t_0) = x_0$.

Since F is continuous, the solution in either case will automatically be C^1 in its domain.

Proof. First, we show that (ii) implies (i). Take any $\varepsilon > 0$ such that $2\varepsilon \leq R/(\sup_{[a, b] \times \overline{B_R(x_0)}} \|F\|)$. Then,

$$\sup_{([t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b]) \times \overline{B_R(x_0)}} \|F\| \leq \sup_{[a, b] \times \overline{B_R(x_0)}} \|F\| \leq \frac{R}{2\varepsilon} \leq \frac{R}{b - a}$$

so we can apply (ii) to get (i).

Now, we prove (ii). Let $X = C([a, b], \overline{B_R(x_0)}) = \{f : [a, b] \rightarrow \overline{B_R(x_0)} \text{ such that } f \text{ is continuous}\}$. This is a larger space than differentiable functions, but we can look for solutions here and then prove that it satisfies the properties we want. X is complete with respect to the uniform metric $d(f, g) = \|f - g\|_\infty$ since X is a closed subset of the complete space $(C([a, b], \mathbb{R}^n), \|\cdot\|_\infty)$. Define, for any $f \in X$, $T(f) : [a, b] \rightarrow \mathbb{R}^n$ with

$$(T(f))(t) = x_0 + \int_{t_0}^t F(s, f(s)) ds.$$

Then, $T(f)$ is continuous. Also,

$$\begin{aligned} \|(T(f))(t) - x_0\| &= \left\| \int_{t_0}^t F(s, f(s)) ds \right\| \\ &\leq \left| \int_{t_0}^t \|F(s, f(s))\| ds \right| \\ &\leq \left| (t - t_0) \sup_{[a, b]} \|F(s, f(s))\| \right| \\ &\leq |t - t_0| \frac{R}{b - a} \\ &\leq R \end{aligned}$$

so $T : X \rightarrow X$. Moreover,

$$\begin{aligned} \|(T(f_1))(t) - (T(f_2))(t)\| &\leq \int_{t_0}^t \|F(s, f_1(s)) - F(s, f_2(s))\| ds \\ &\leq K|t - t_0| \|f_1 - f_2\|_\infty \\ &\leq K(b - a) \|f_1 - f_2\|_\infty. \end{aligned}$$

If $K(b - a) < 1$, then $T : X \rightarrow X$ is a contraction. By the contraction mapping theorem, there is a unique $f \in X$ such that $f(t) = x_0 + \int_{t_0}^t F(s, f(s)) ds$. So, $f(t_0) = x_0$. Also, by the fundamental theorem of calculus, f

is differentiable with $df/dt = F(t, f(t))$. In general, for $K(b-a) \geq 1$, we argue that $T^{(m)}(f) = \underbrace{T \circ \dots \circ T}_{m \text{ times}}(f)$ is a contraction $X \rightarrow X$.

Lemma 5.1

For any $t \in [a, b]$, we have

$$\left\| (T^{(m)}(f_1))(t) - (T^{(m)}(f_2))(t) \right\| \leq \frac{|t - t_0|^m K^m}{m!} \|f_1 - f_2\|_\infty.$$

Proof. We proceed by induction on m . We already proved the case $m = 1$. For $m \geq 2$ and $t \geq t_0$,

$$\begin{aligned} \left\| (T^{(m)}(f_1))(t) - (T^{(m)}(f_2))(t) \right\| &\leq K \int_{t_0}^t \left\| (T^{(m-1)}(f_1))(s) - (T^{(m-1)}(f_2))(s) \right\| ds \\ &\leq K \|f_1 - f_2\|_\infty \frac{K^{m-1}}{(m-1)!} \int_{t_0}^t (s - t_0)^{m-1} ds \\ &\leq \frac{|t - t_0|^m K^m}{m!} \|f_1 - f_2\|_\infty. \end{aligned}$$

□

So,

$$\left\| T^{(m)}(f_1) - T^{(m)}(f_2) \right\|_\infty \leq \frac{(b-a)^m K^m}{m!} \|f_1 - f_2\|_\infty \rightarrow 0 \text{ as } m \rightarrow \infty.$$

So, for large enough m , $T^{(m)} : X \rightarrow X$ is a contraction. Hence, by the following variant of the contraction mapping theorem, T has a unique fixed point $f \in X$.

Corollary 5.1

Let $X \neq \emptyset$ be complete and $T : X \rightarrow X$. If $T^{(m)}$ is a contraction, then T has a unique fixed point.

Proof. $T^{(m)}$ has a unique fixed point $x \in X$ by the usual contraction mapping theorem. $T^{(m)}(x) = x$ so $T^{(m)}(T(x)) = T^{(m+1)}(x) = T(x)$. That is, $T(x)$ is also a fixed point of $T^{(m)}$. So, by uniqueness, $T(x) = x$. If $T(y) = y$, then $T^{(m)}(y) = y$ so again by uniqueness $y = x$. □

And we are done. □